

Anatomy & Physiology of the Vestibular System



PPOM1

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Session Objectives

- Describe the anatomy of the peripheral vestibular system including the otolith organs and semicircular canals.
- Describe the anatomy and physiology of the hair cells in the maculae and cristae.
- Describe central vestibular system anatomy and its role in postural and oculomotor control.
- Describe the pathology of the vestibular system disorders such as BPPV among others presented.

Vestibular system helps to maintain posture, balance, and eye fixation as we move in the environment. The vestibular system is pushed to the limit in our modern lifestyles.

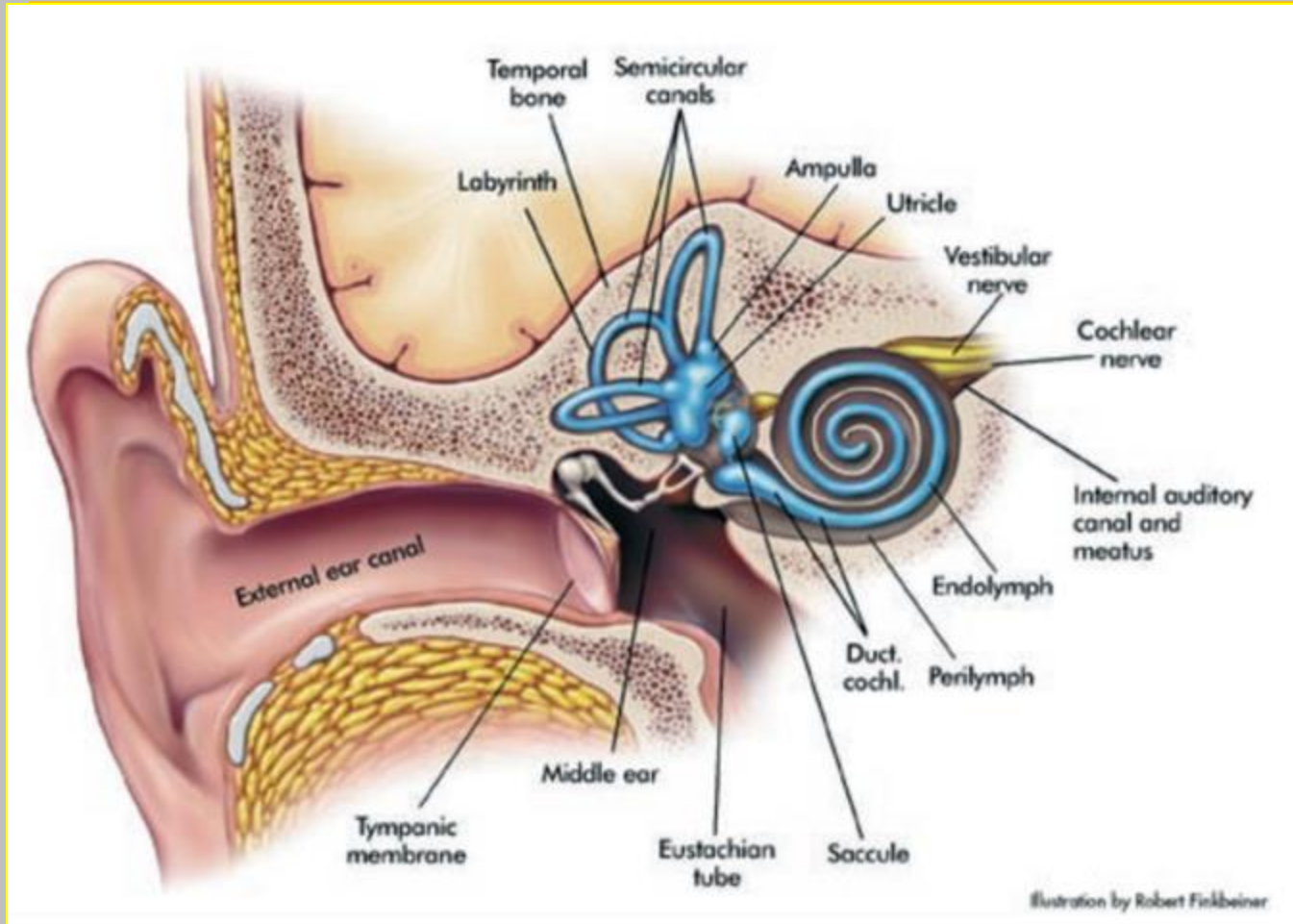
Vestibular dysfunction generally presents with vertigo (sensation of world moving when it is not) and postural instability.

Clinical signs include nystagmus (repetitive, uncontrolled eye movements) and positive Romberg tests.

Vestibular disorders can be peripheral or central.



Anatomy of the Peripheral Vestibular System

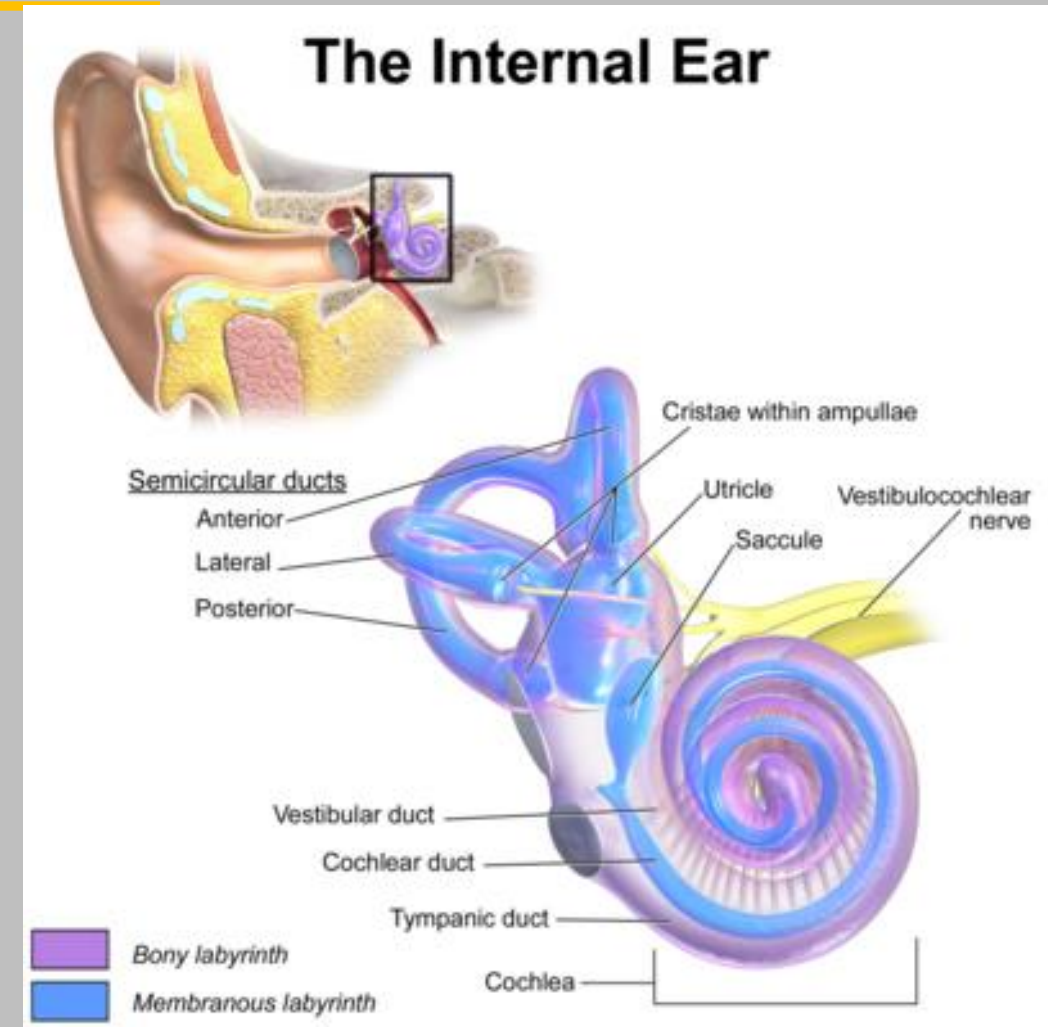


Otolith organs
- sacule
- utricle

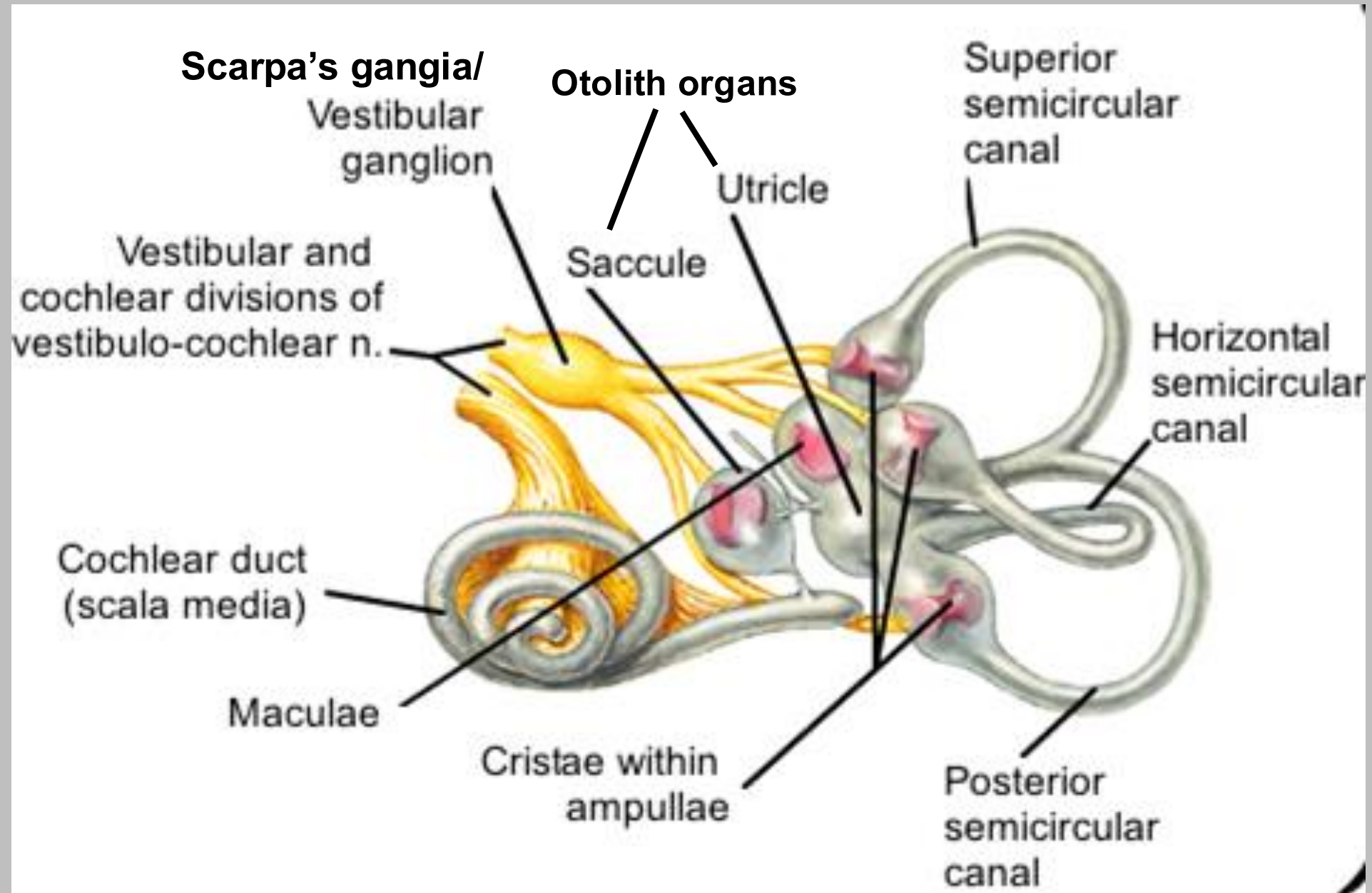
Semicircular canals

Endolymph vs Perilymph

Vestibular portion of CNVIII



Anatomy of the Peripheral Vestibular System



Otolith organs

- Saccule & Utricle are cavities filled with endolymph

- Each has a macula lined with hair cells

- covered by a loose otolithic membrane which is high in specific gravity due to its gelatinous composition.

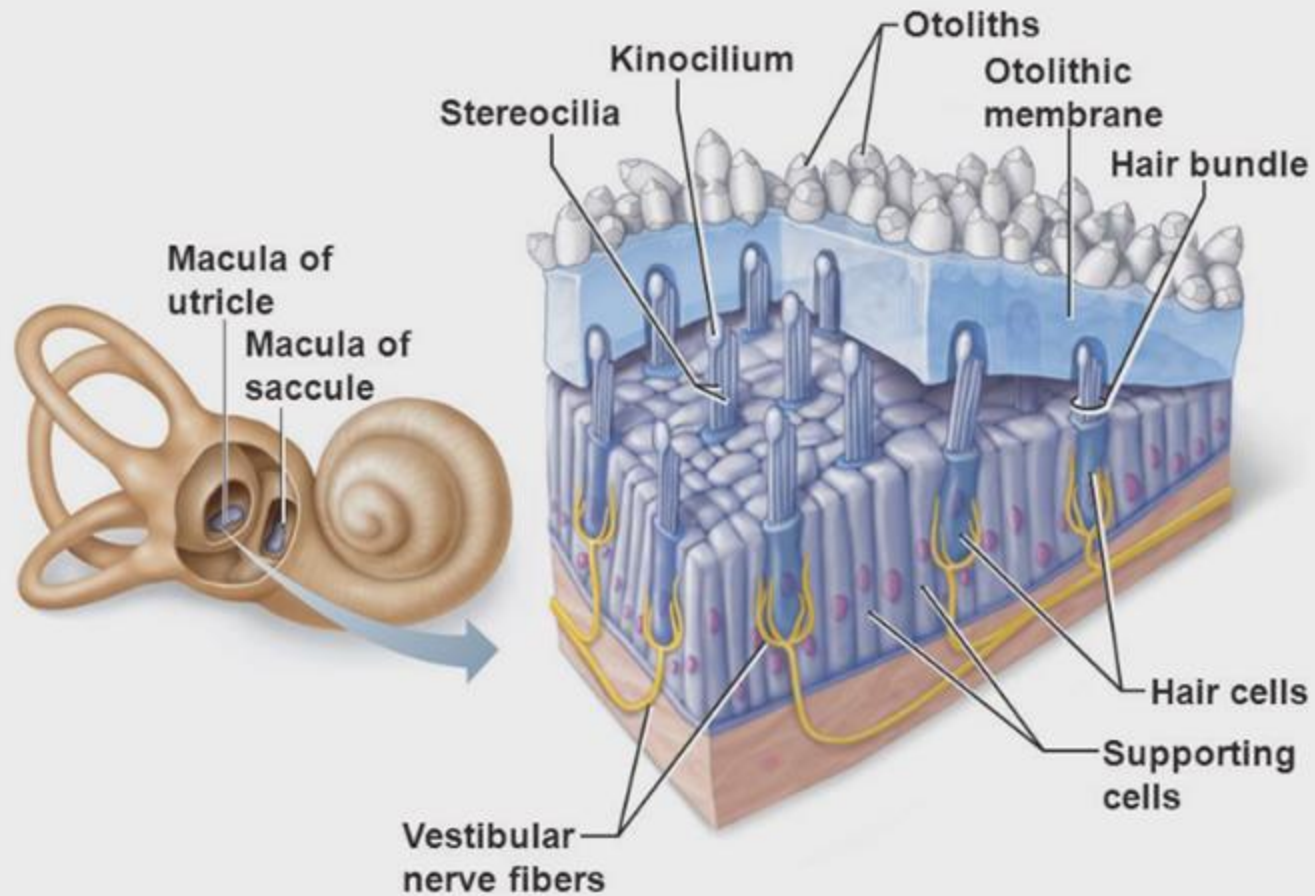


Figure 15.34

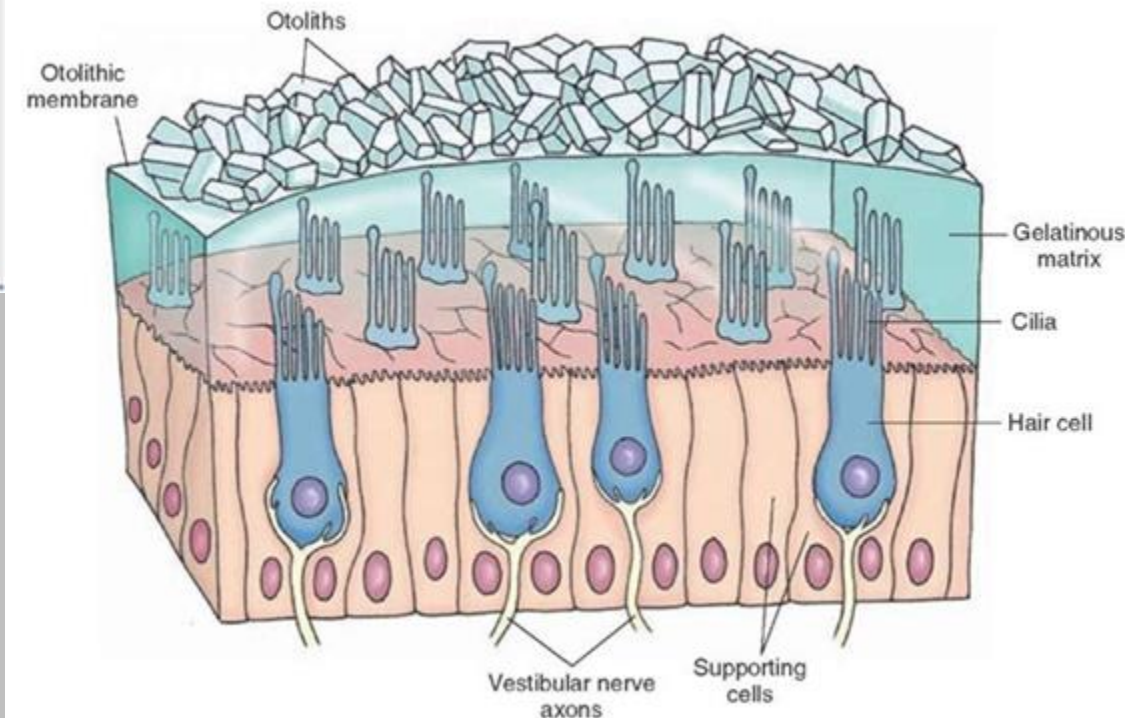
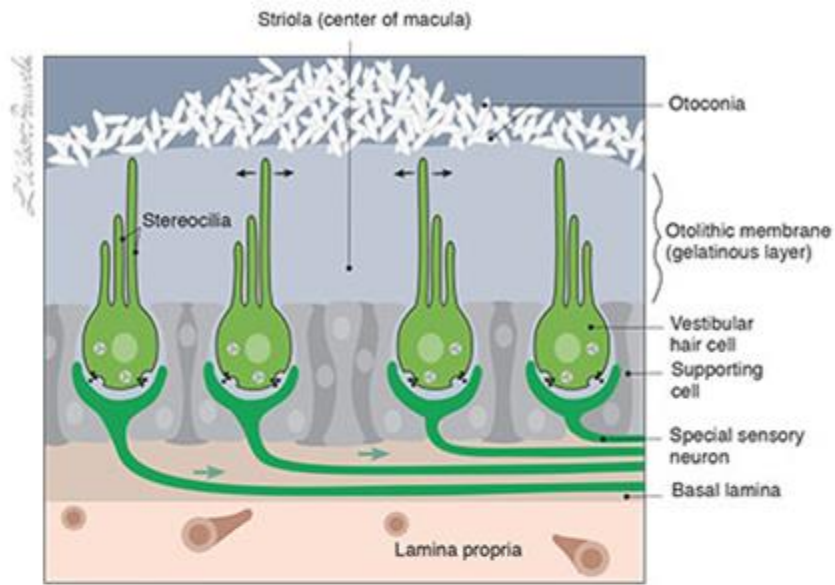
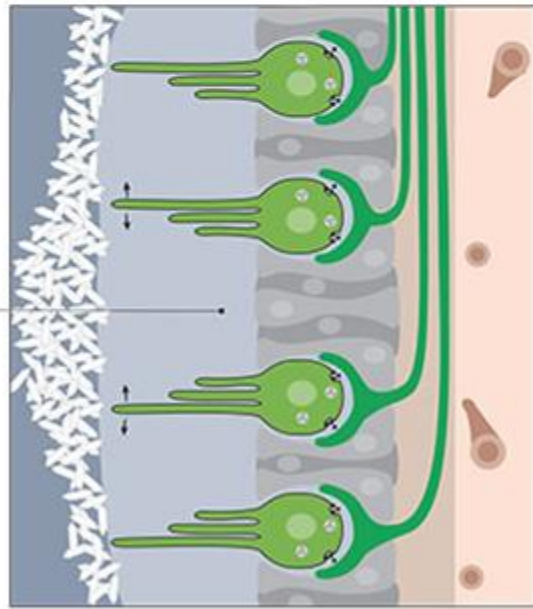


Figure VIII-5 Macular hair cells of the utricle and saccule

From Cranial Nerves 3rd Ed. ©2010 Wilson-Pauwels, Stewart, Akesson, Spacey, PMPH-USA



UTRICLE

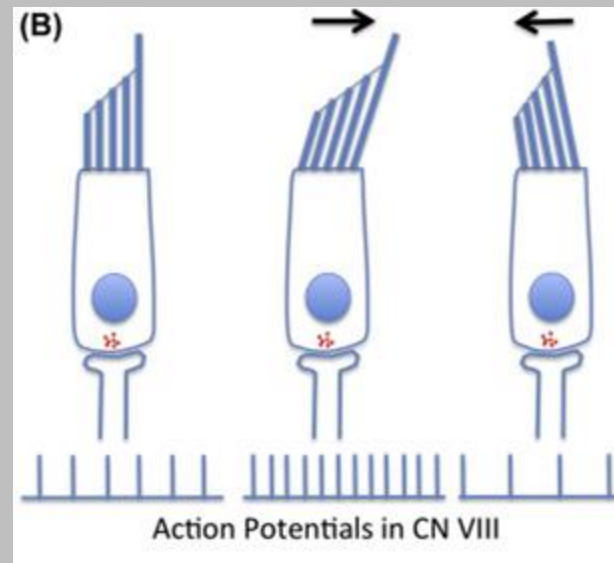
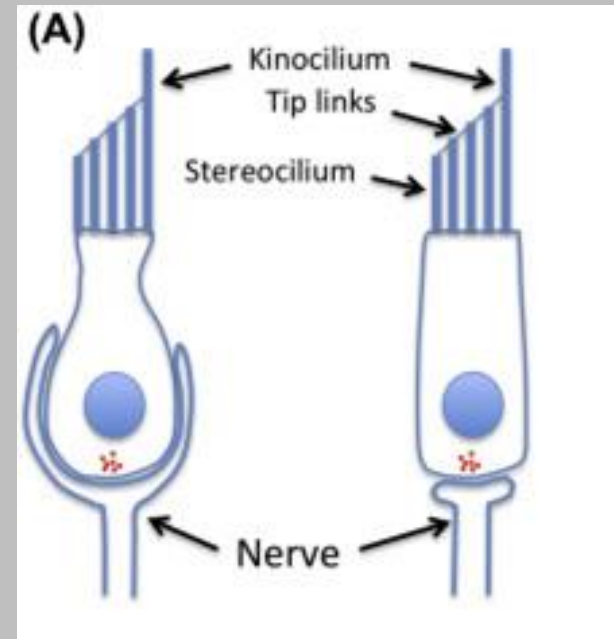


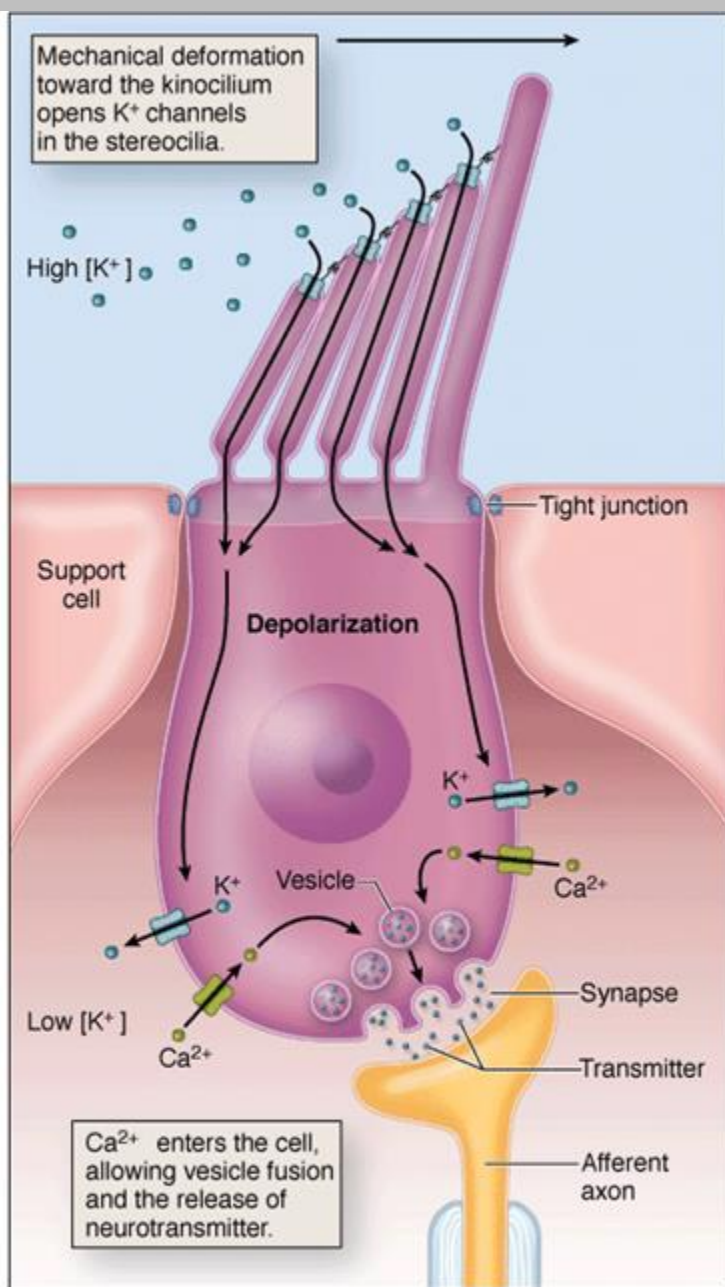
SACCULE

- Utricle – hair cells Upright
- Saccule – hair cells on Side

Encode the position and movement of the head with respect to gravity (linear acceleration) via activity of hair cells (deflection of cilia...tip links...K⁺/Ca⁺... glutamate...).

Bending in one direction causes depolarization.
Bending in opposite direction causes hyperpolarization.





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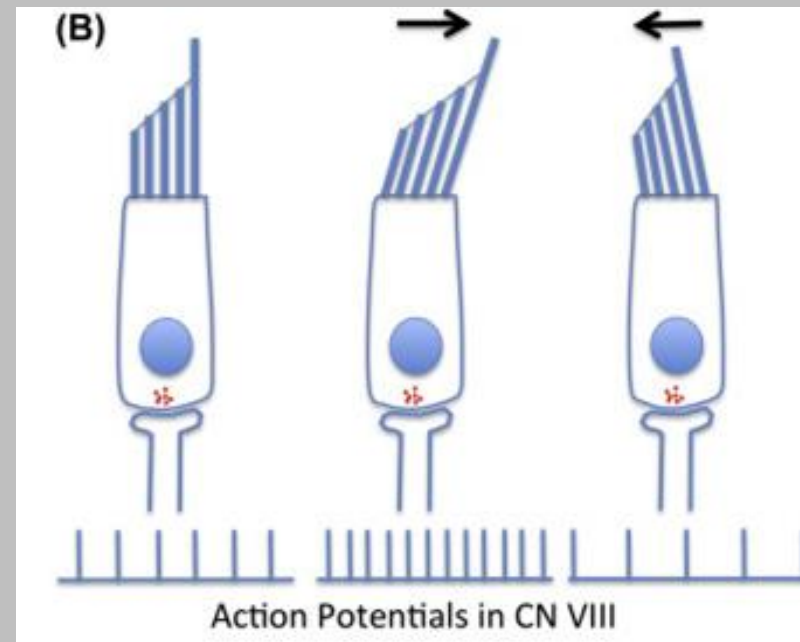
Source: Mescher AL: Junqueira's Basic Histology: Text and Atlas, 12th Edition: <http://www.accessmedicine.com>
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Deformation of hair cell cilia in the direction of the kinocilium results in depolarization of the hair cells via opening of K^+ channels on tip links.

Depolarization causes voltage gated Ca^{++} channels to open resulting in transmitter fusion, exocytosis, and transmitter release.

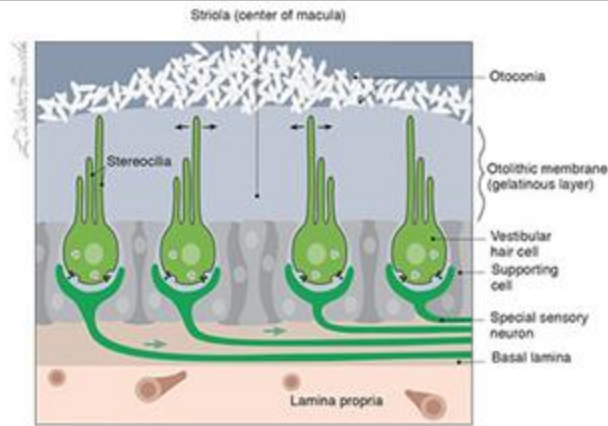
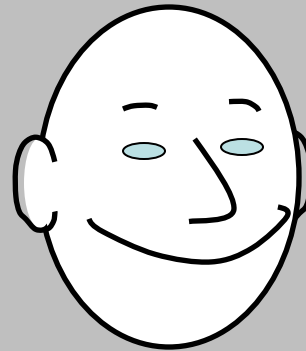
Binding of glutamate onto receptors on CNVIII cells causes Na^+ influx and depolarizes these cells causing action potential generation.

Deformation of hair cell cilia in the direction opposite of the kinocilium results in hyperpolarization.

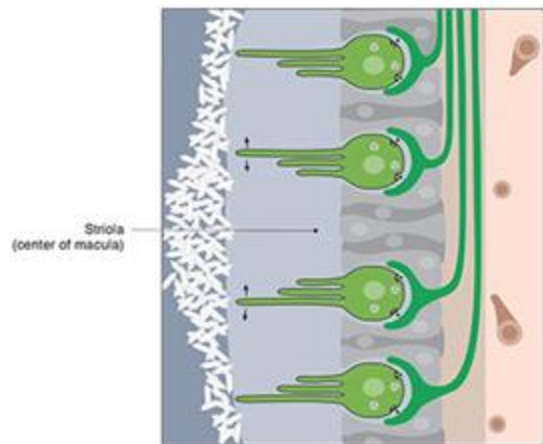


Otolith organs: sensory transduction

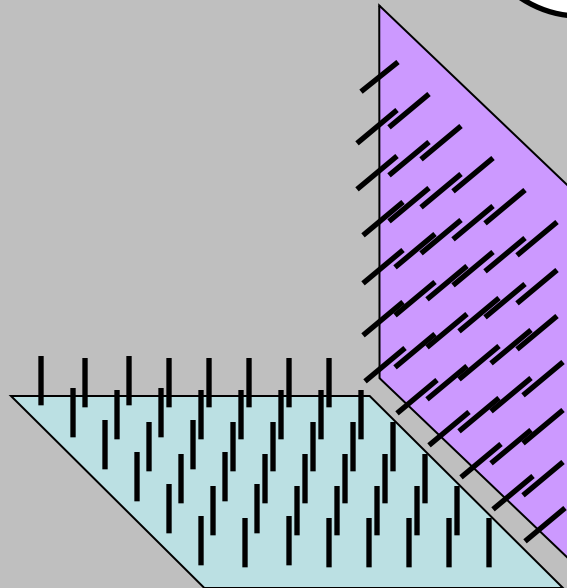
The **OTOLITH** organs are sensitive to gravity and **linear** head motion



UTRICLE



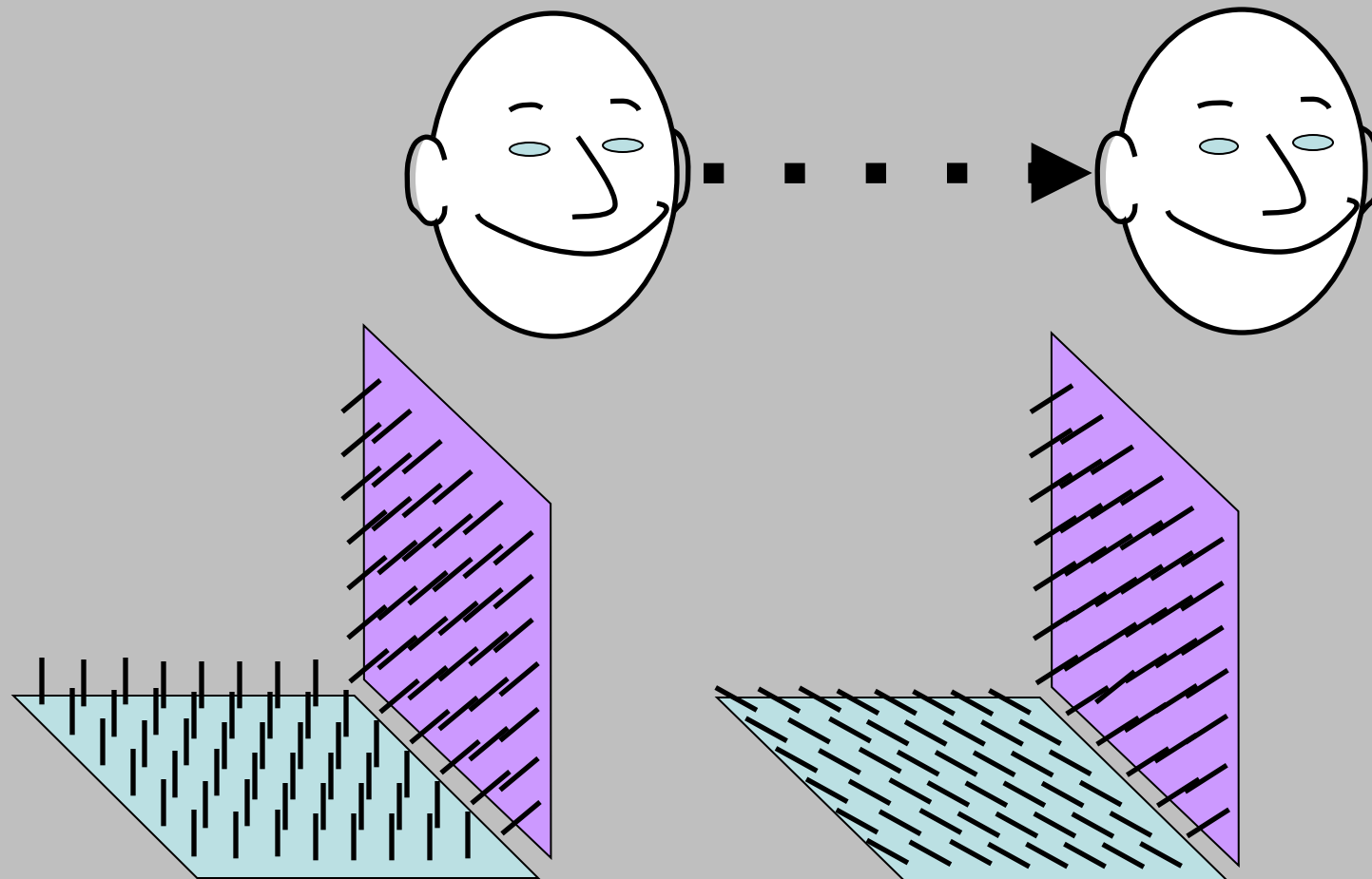
SACCULE



Otolith organs: sensory transduction

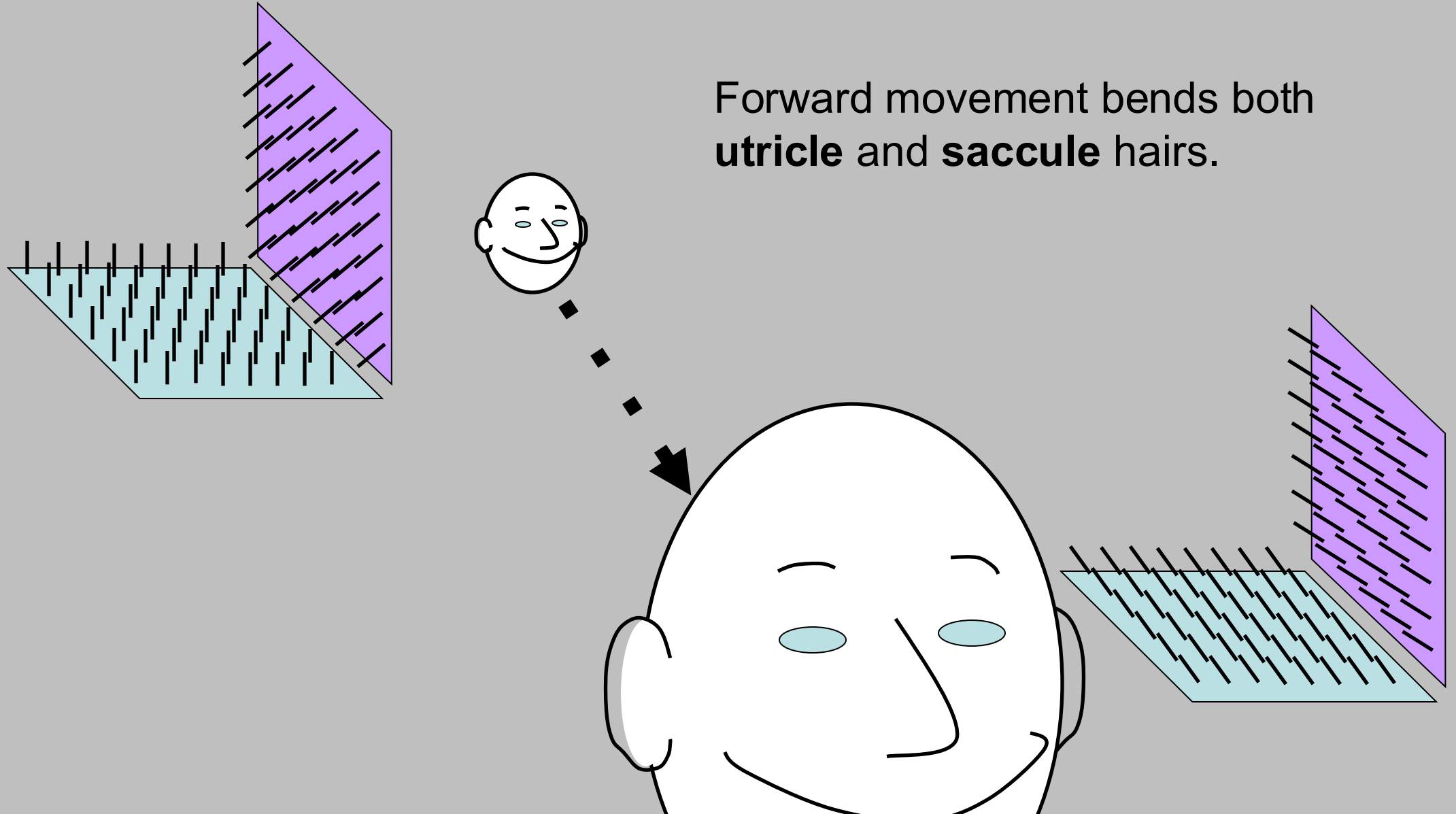
Linear head motion

Lateral horizontal movement mostly bends **utricle** hairs while hair cells in saccule remain unchanged.



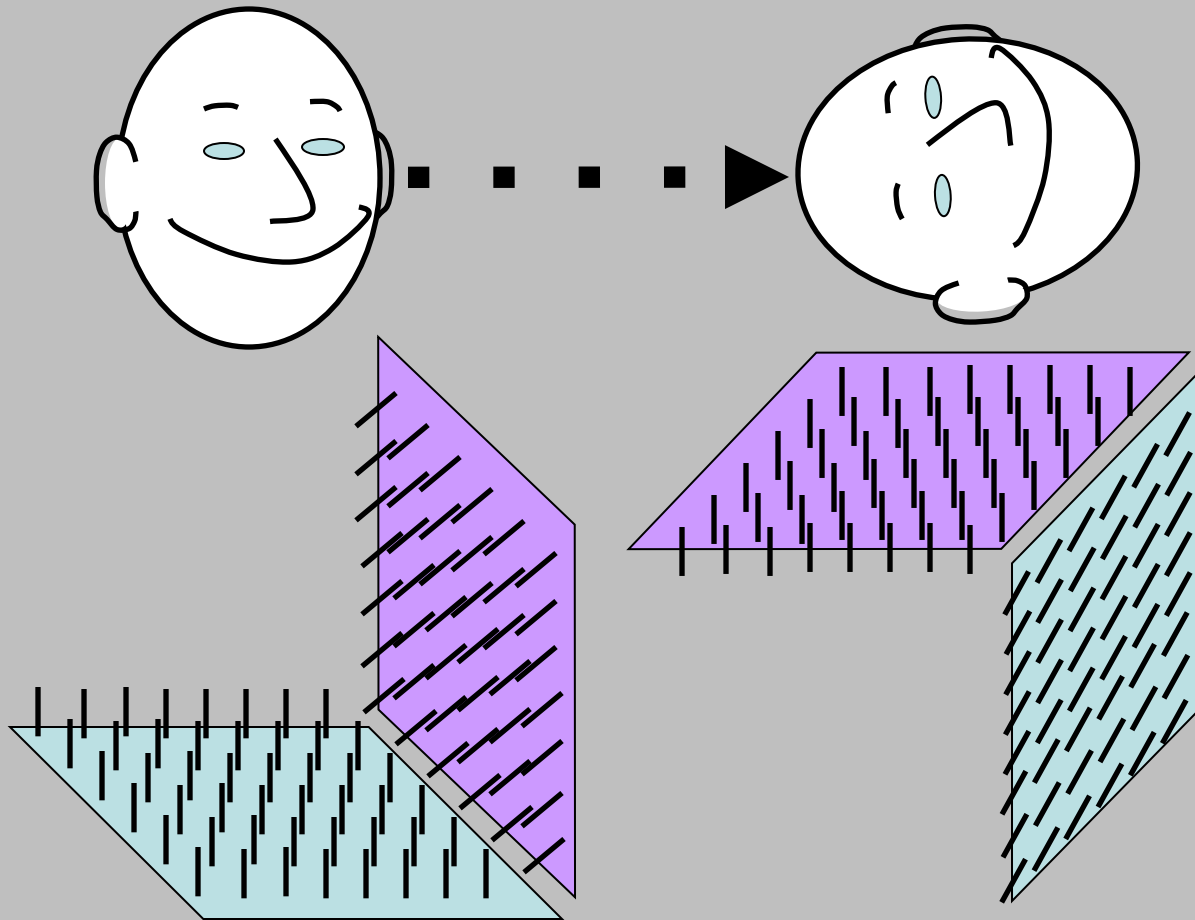
Otolith organs: sensory transduction

Linear head motion



Otolith organs: sensory transduction

Changing head orientation w/ respect to **GRAVITY** will change in hair cell position

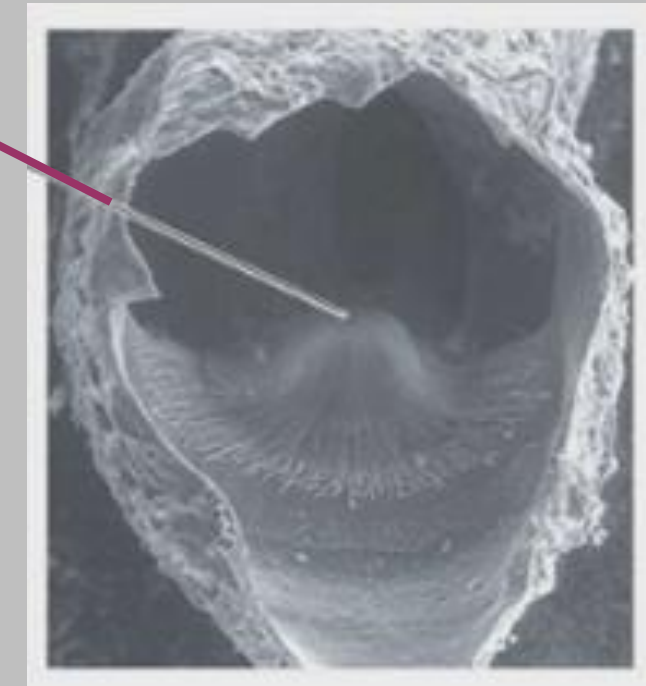
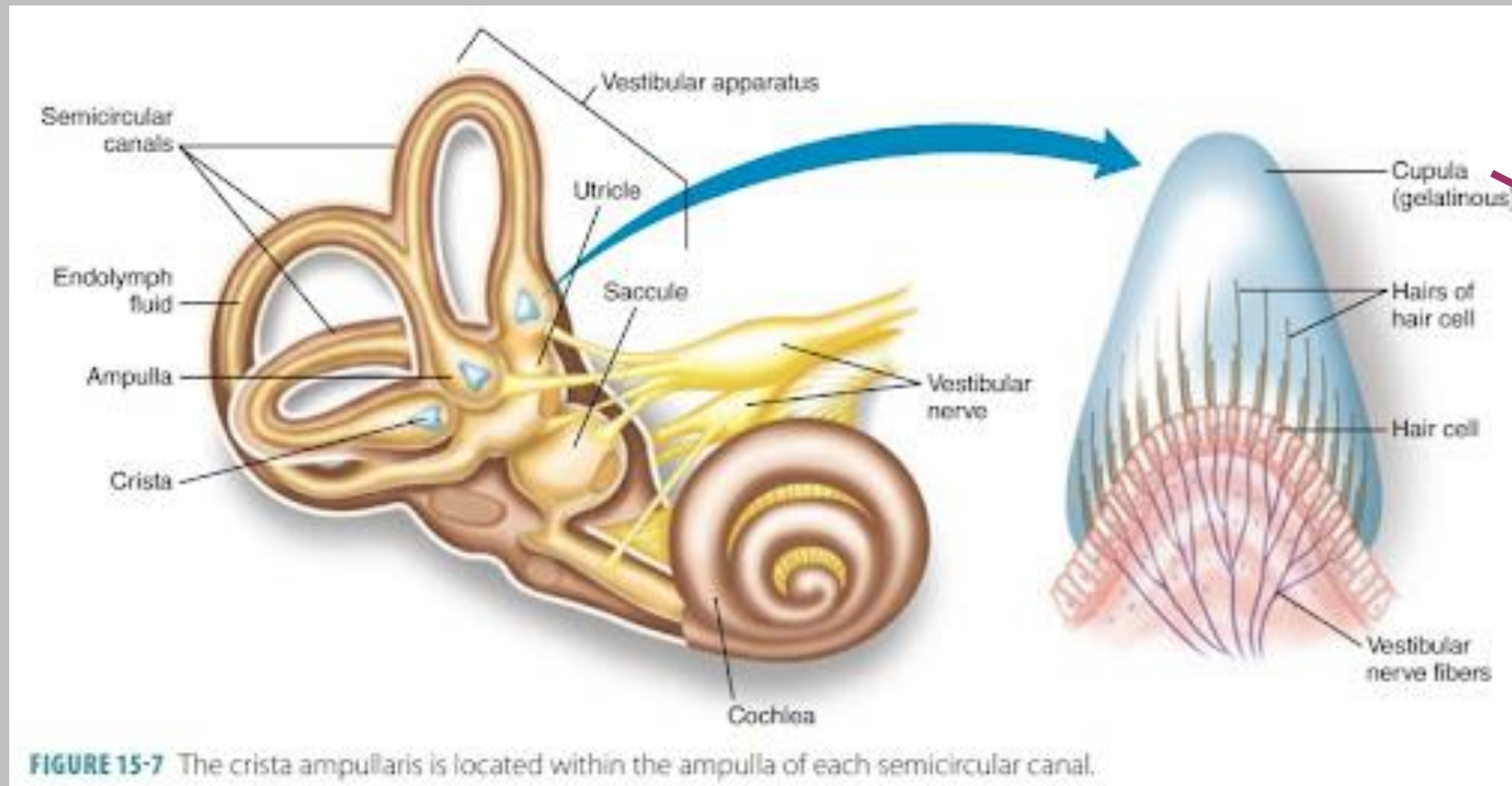
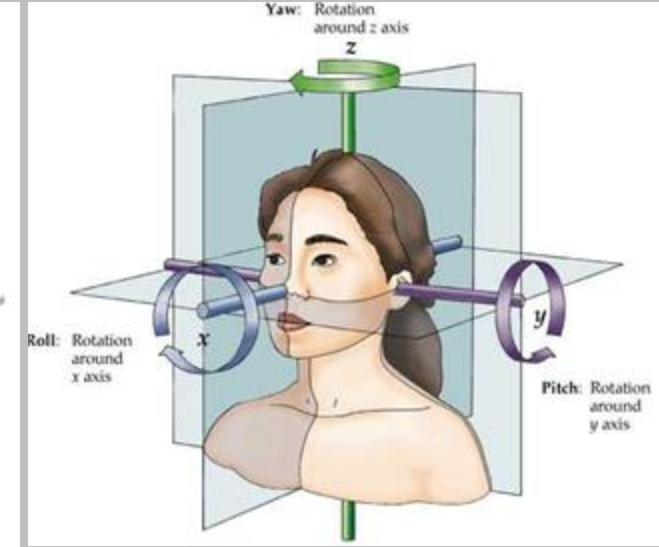
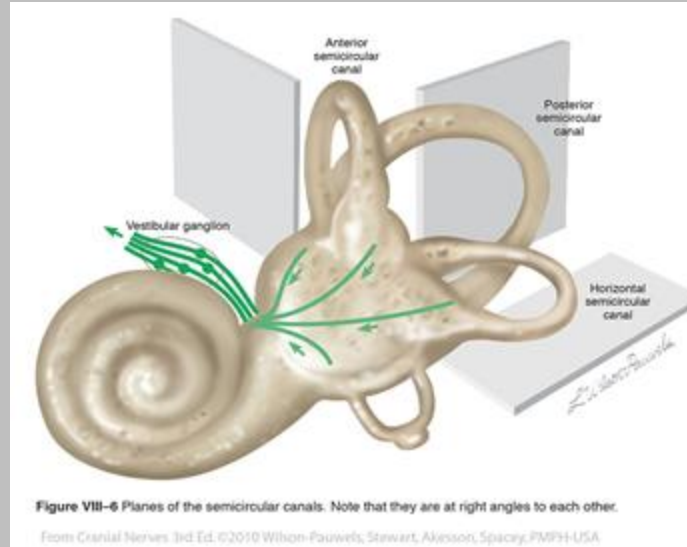


Anatomy of the Semi-circular canals

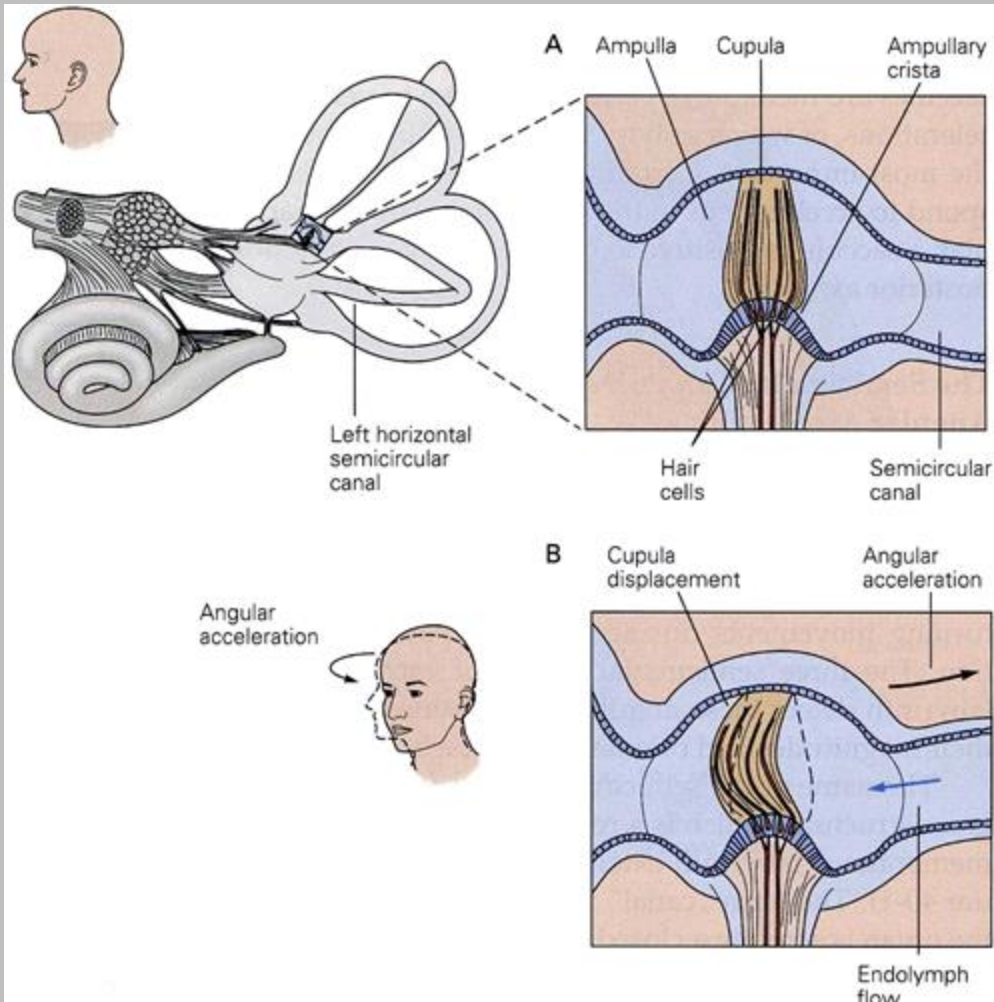
3 ducts emerging from the utricle each oriented along the 3 major planes (x, y, z)

Detects **angular head** movements along the 3 major planes (x, y, z)

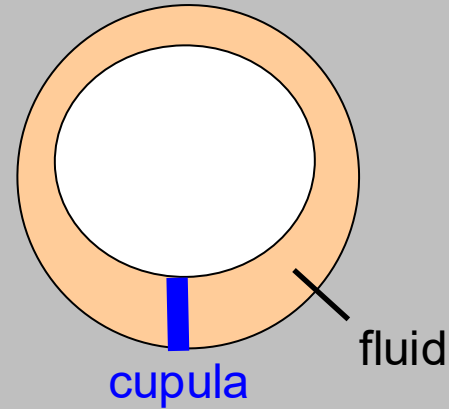
Each contain an ampula where sensory cells are located.



Semicircular canals: sensory transduction-- **angular** head motion



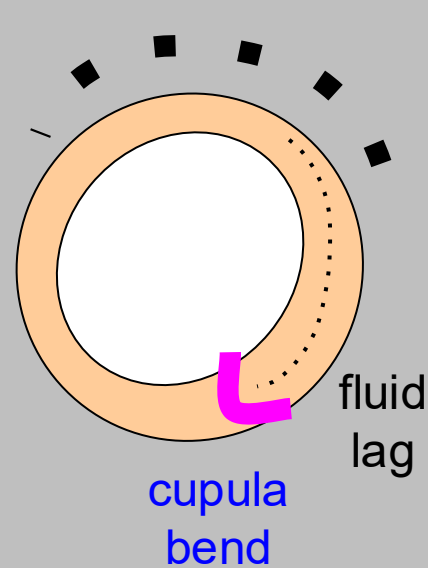
stationary



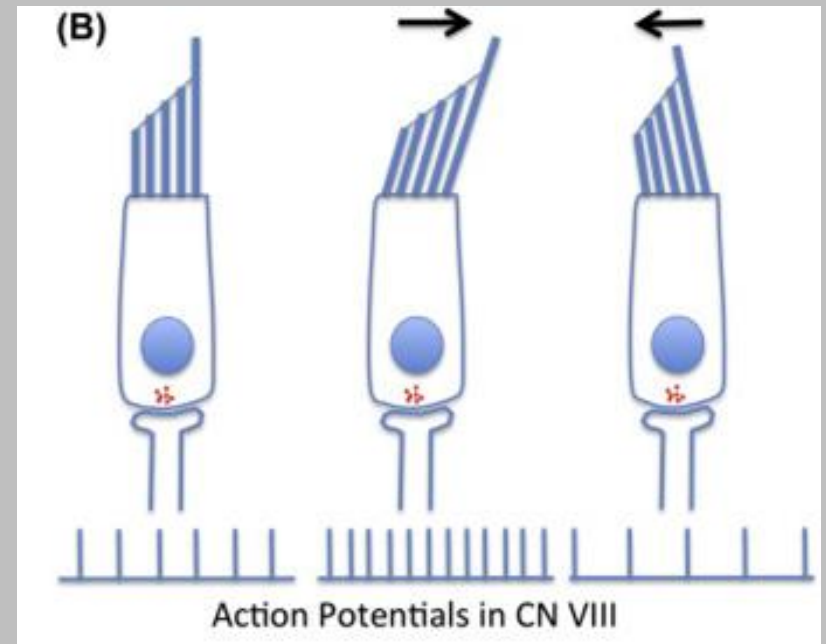
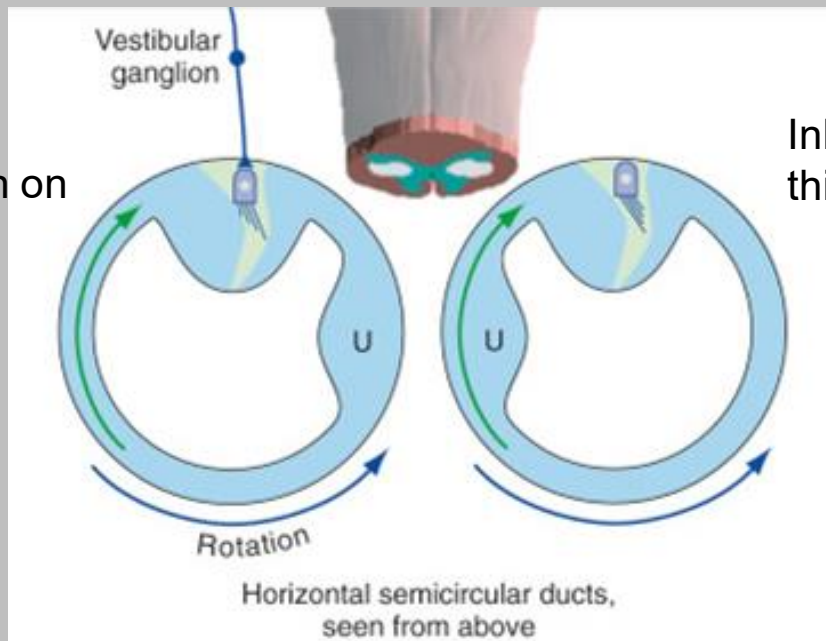
The cristae respond to rotary acceleration

During a rotation, the endolymph lags behind the head acceleration which causes deflection of the cupola opposite to the head movement

head turn



Hair cells are deflected causing a change in their membrane potential causing more or less transmitter release onto CNVIII cells.



The cristae respond to rotary acceleration

Endolymph lags the head acceleration deflecting the cupola opposite to the head movement

Hairs and cilia in cupula bend

The largest cilia are arranged toward the midline

Excitation occurs when the flow causes bending of cilia toward midline.

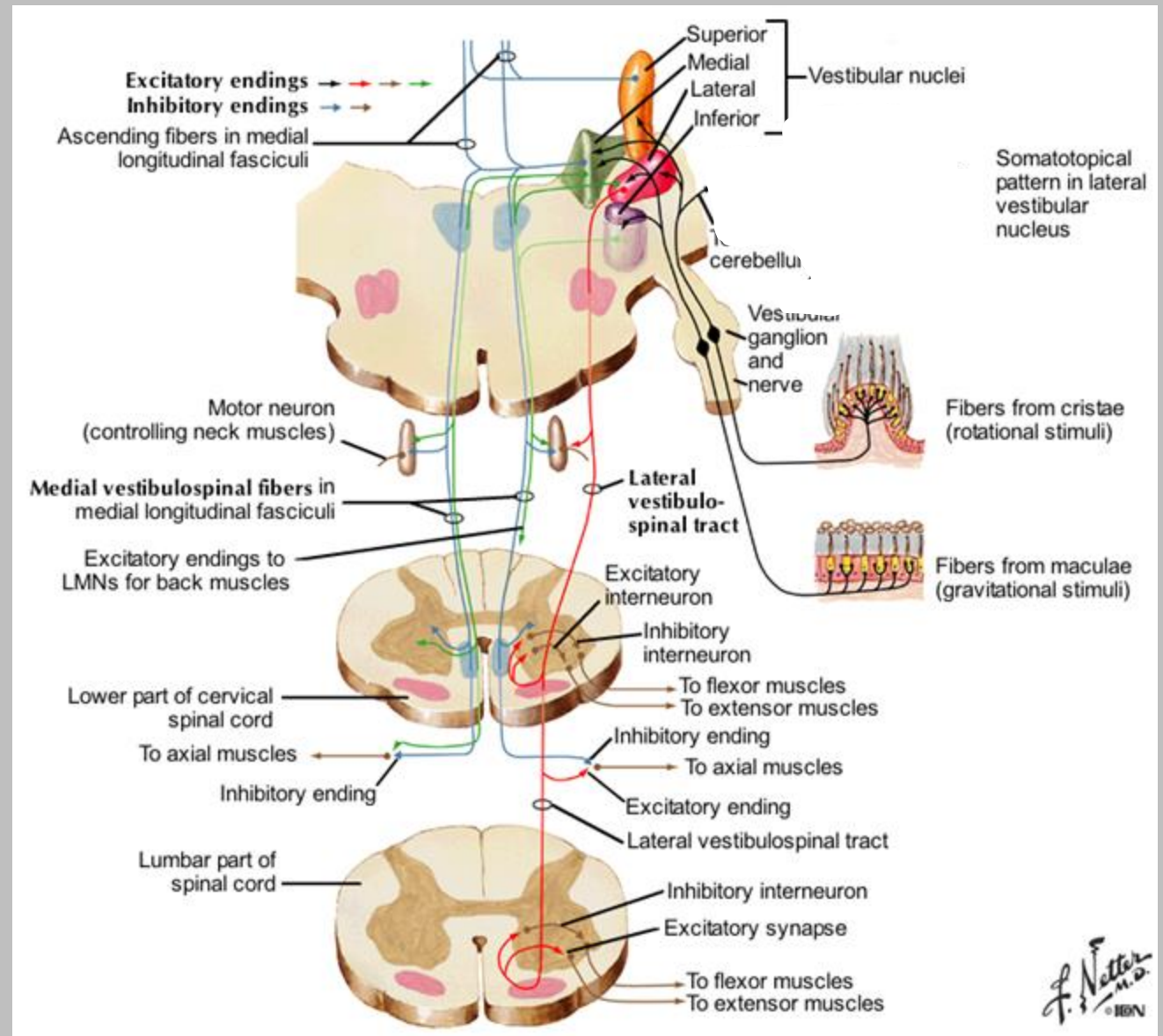
Inhibition occurs when the flow causes bending of cilia toward lateral

The hair cells synapse to the peripheral process of cells in the vestibular ganglion

Vestibular pathways:

Vestibulo-spinal pathways controls corrective actions during movement to maintain posture.

- Lateral vestibulo-spinal tract
 - **Lateral and inferior vestibular nuclei ipsilateral projection.** Project onto anterior horn and innervate proximal muscles of limb **contraction of forelimb and hindlimb muscles** to maintain balance and upright posture
- Medial vestibulo-spinal tract
 - **Medial vestibular nucleus project bilaterally** (medial longitudinal fasciculus) innervate neck muscles including SCM, **stabilization of head position**



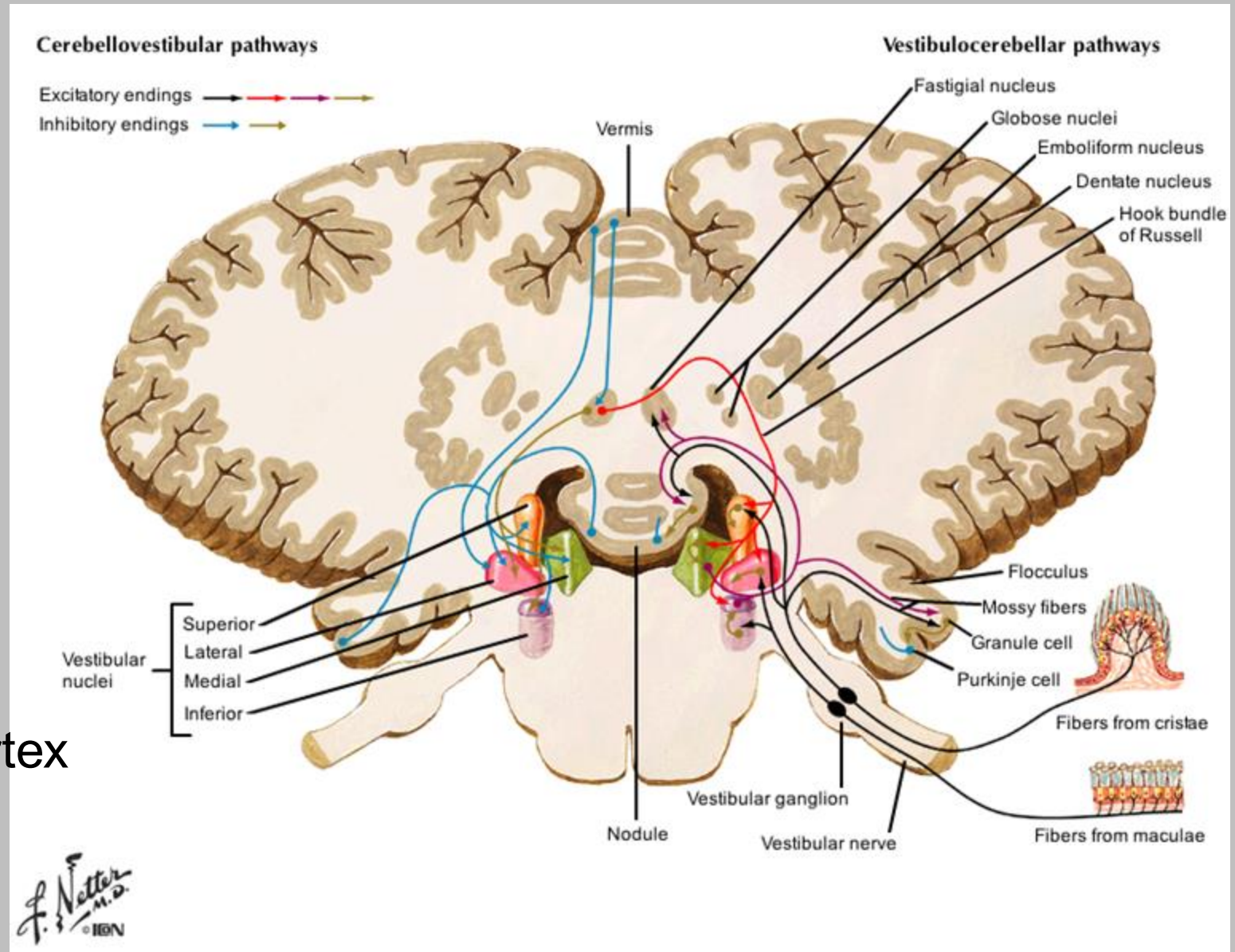
Vestibular pathways:

2] Vestibulo-cerebellar pathway

Controls corrective actions during movement to maintain posture.

Includes projections to deep cerebellar nuclei & cerebellar cortex (flocculus & nodule)

Includes only Ipsi projections

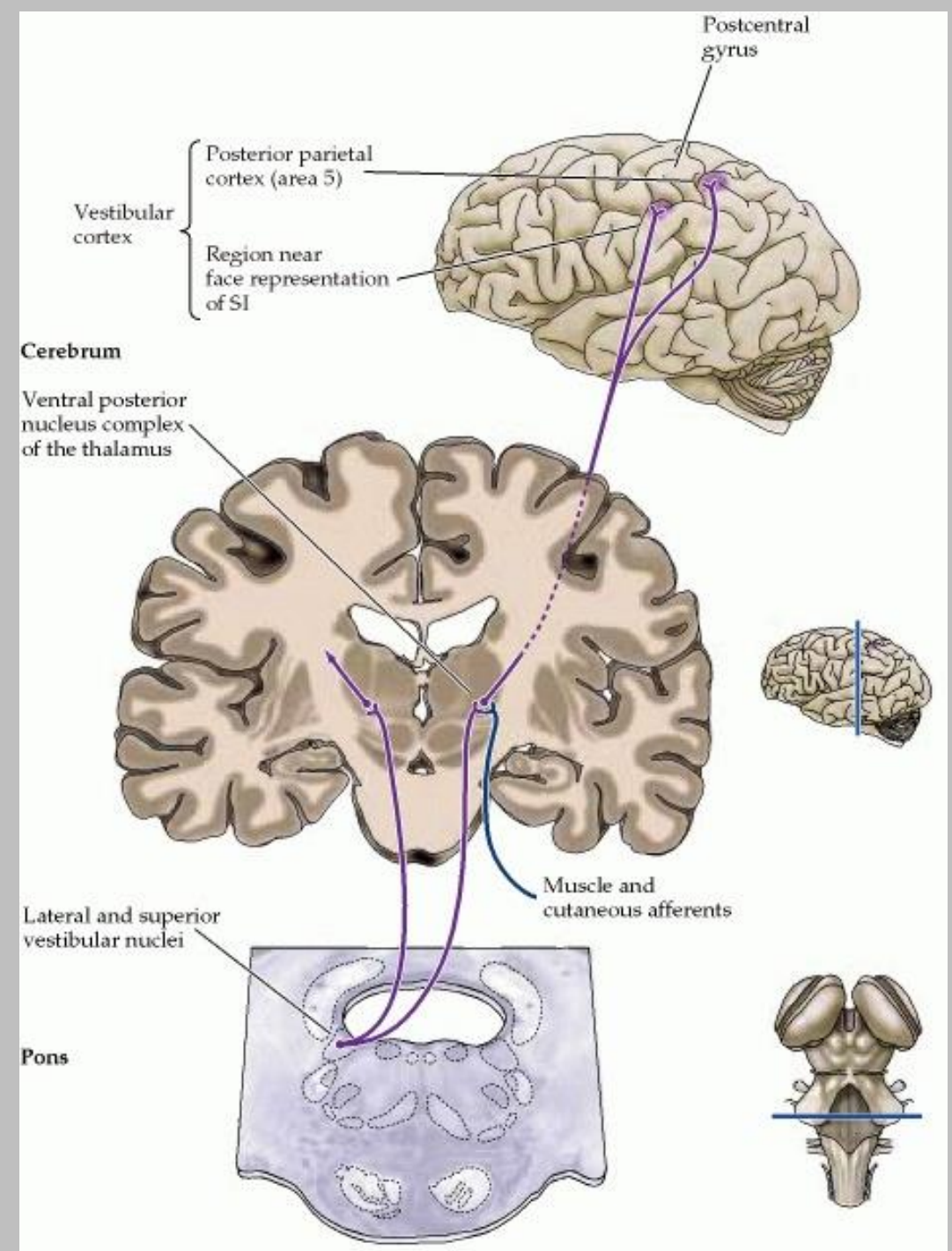


Vestibular pathways:

3] Vestibulo-thalamic pathway

Essential for conscious perception of body and eye position/movement in relation to environment

Includes Ipsi & Contra projections

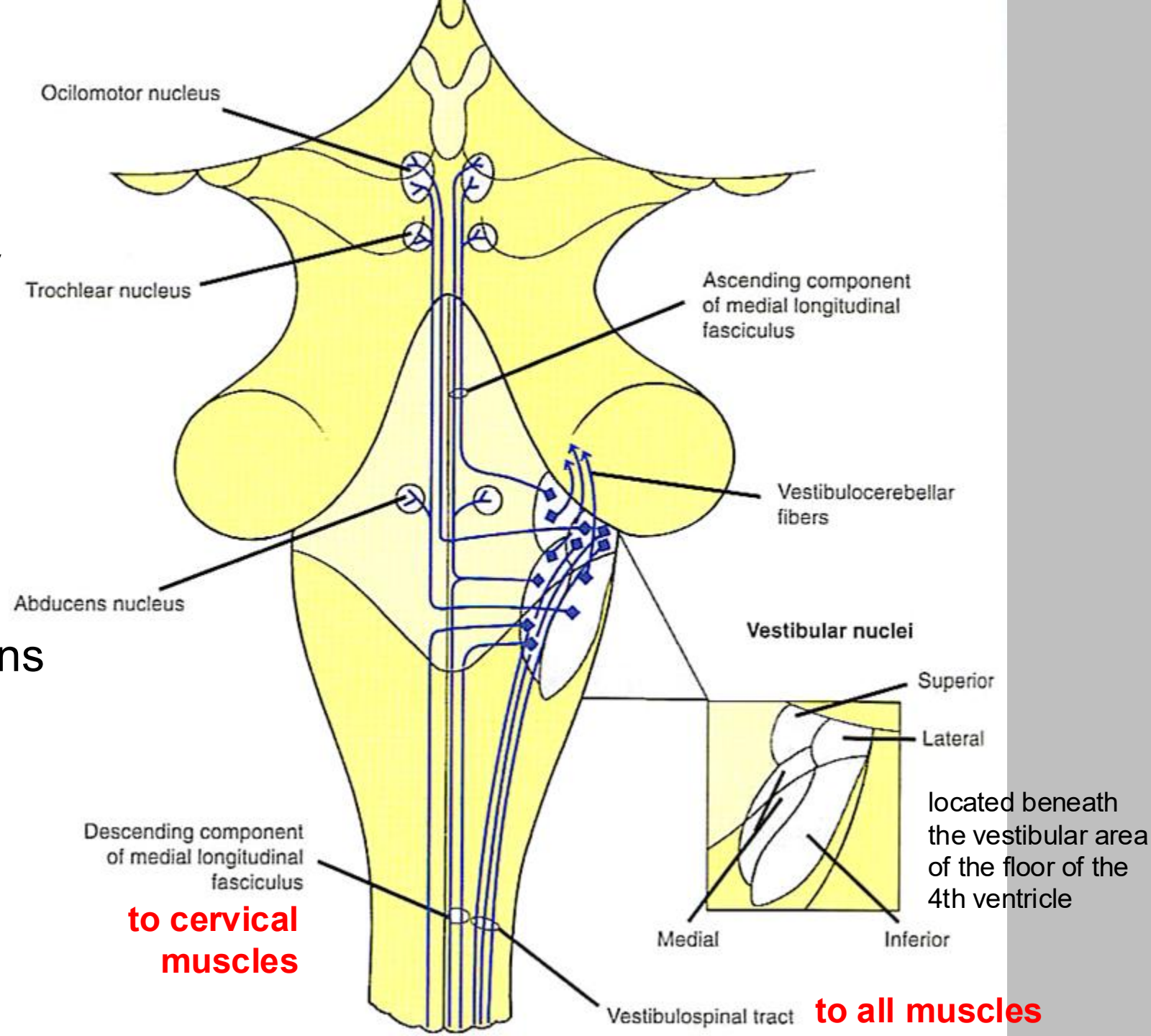


Vestibular pathways:

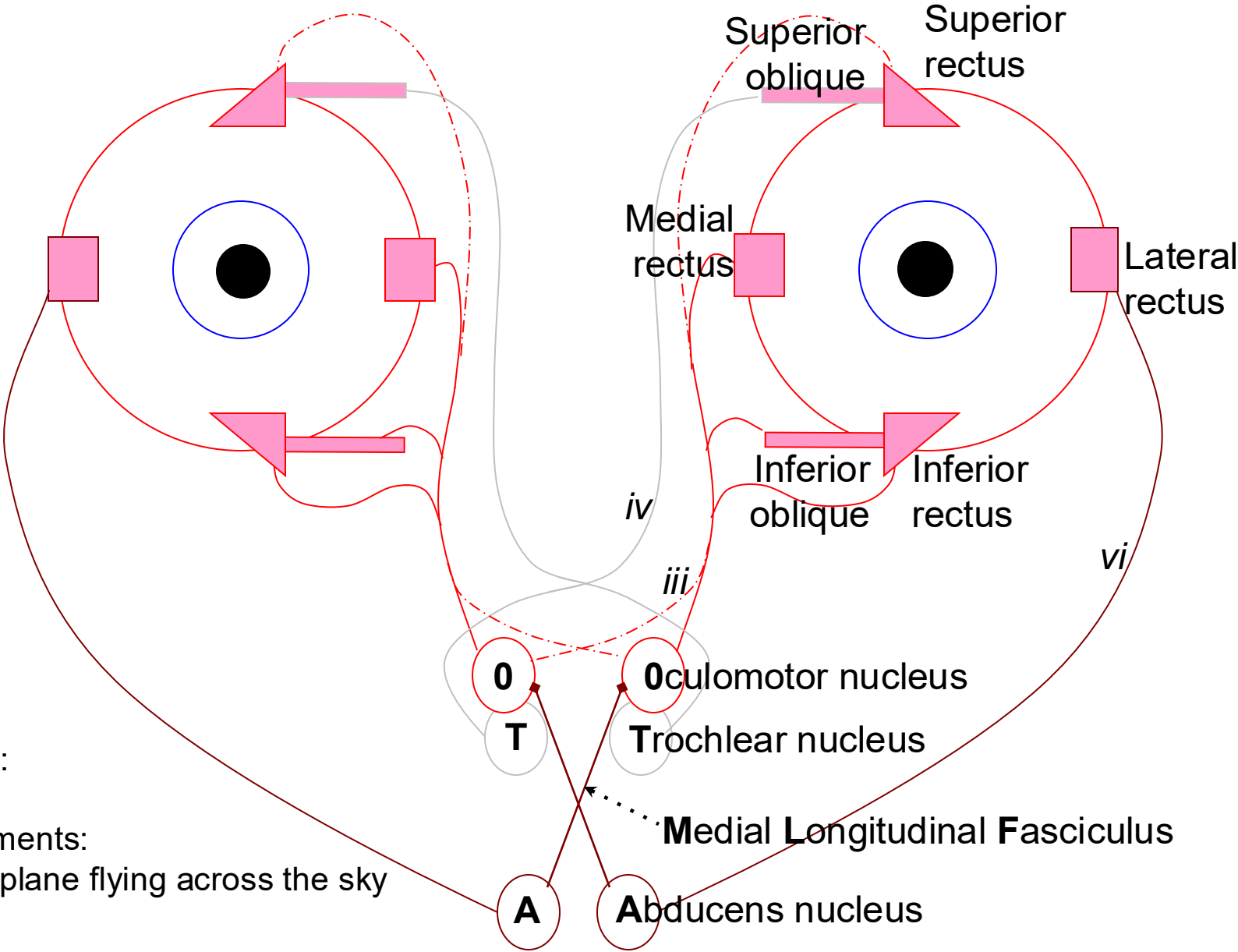
4] Vestibulo-oculo pathway

Controls corrective eye movements to maintain posture and fixation during movement.

Includes Ipsi & Contra projections

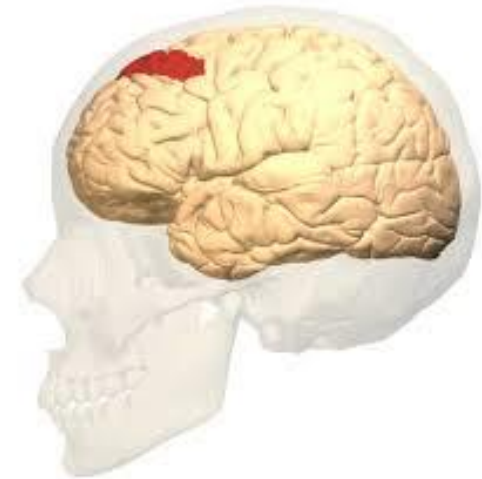
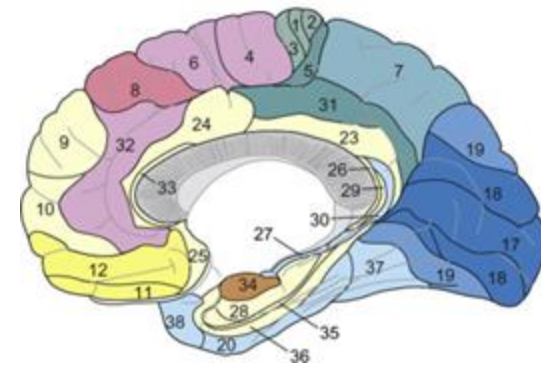
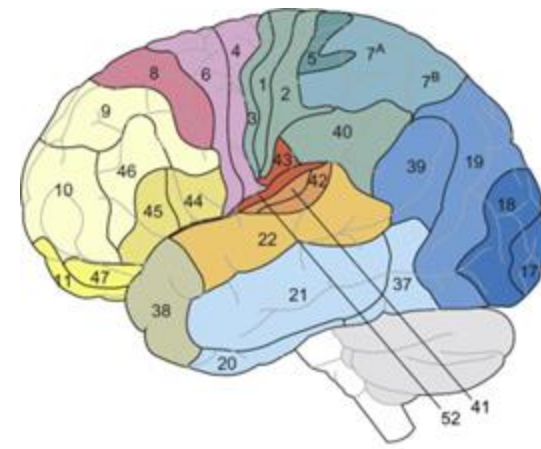
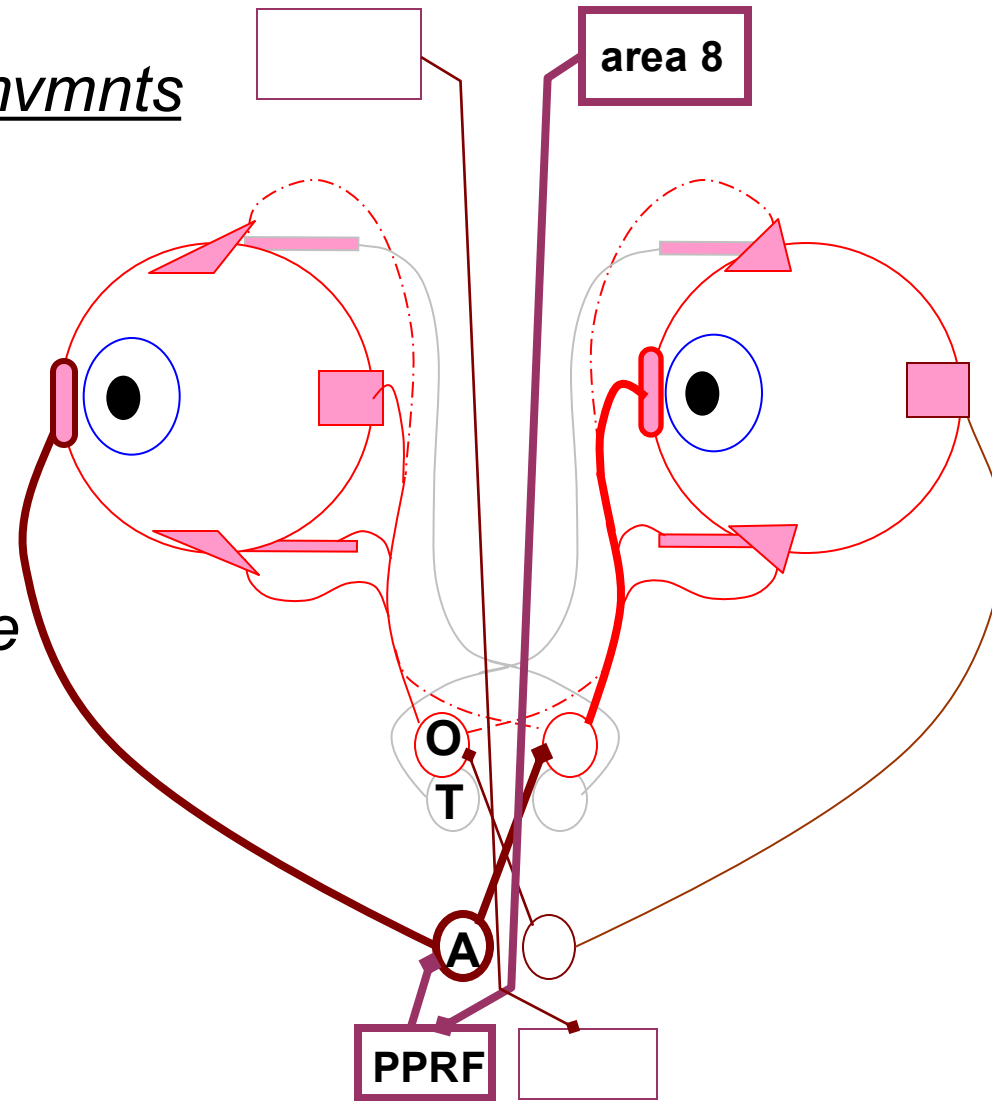


CN III, IV, VI control eye movements

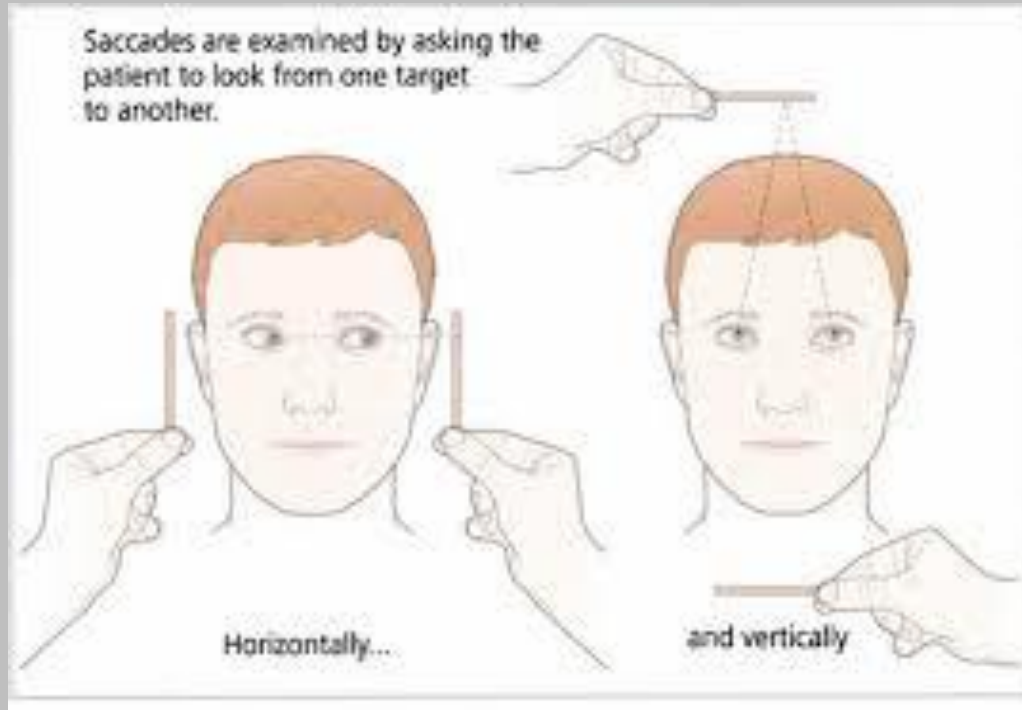


Voluntary lateral eye mvmnts

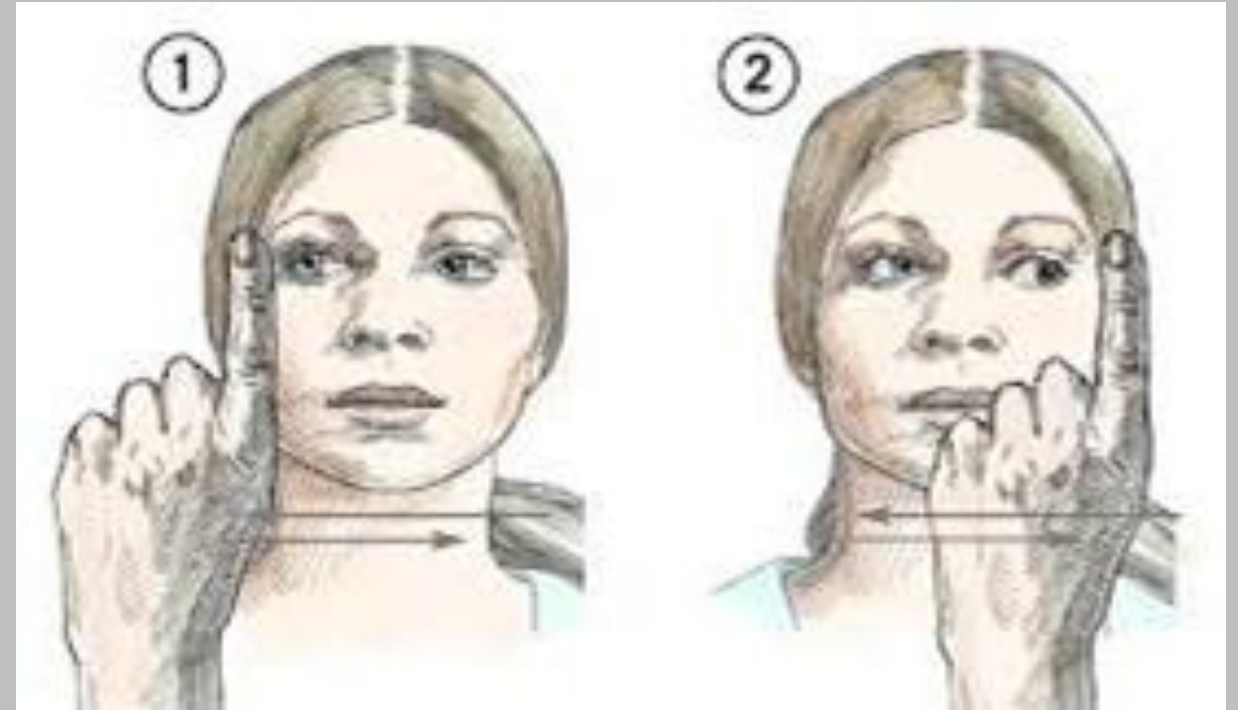
*originates in
cortical **area 8**,
the frontal eye field
and is mediated by
the **P**aramedian **P**ontine
Reticular **F**ormation
(PPRF)*



Clinical testing of **Saccadic** eye movements



Clinical testing of **Smooth pursuit** eye movements

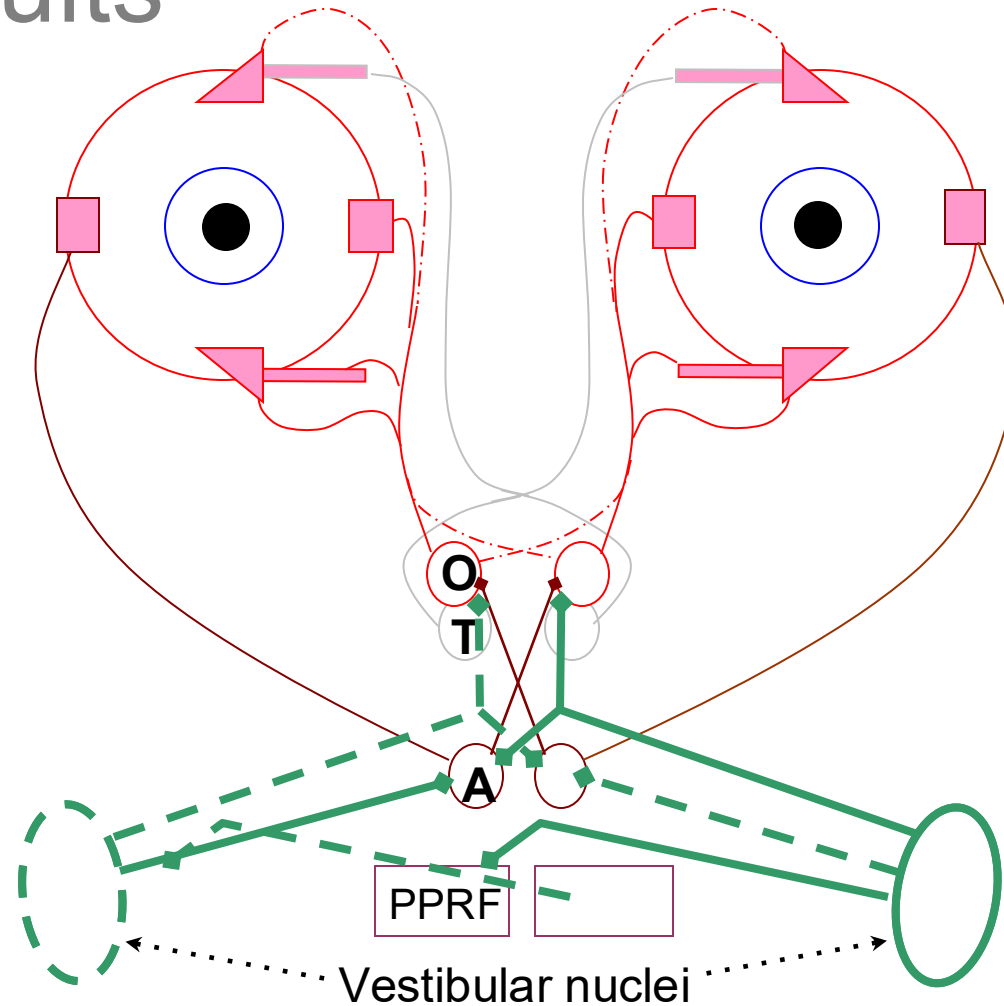


These eye movements are voluntary and do not engage the vestibular system when there is no movement of the head or body

Vestibular system regulation of oculomotor circuits

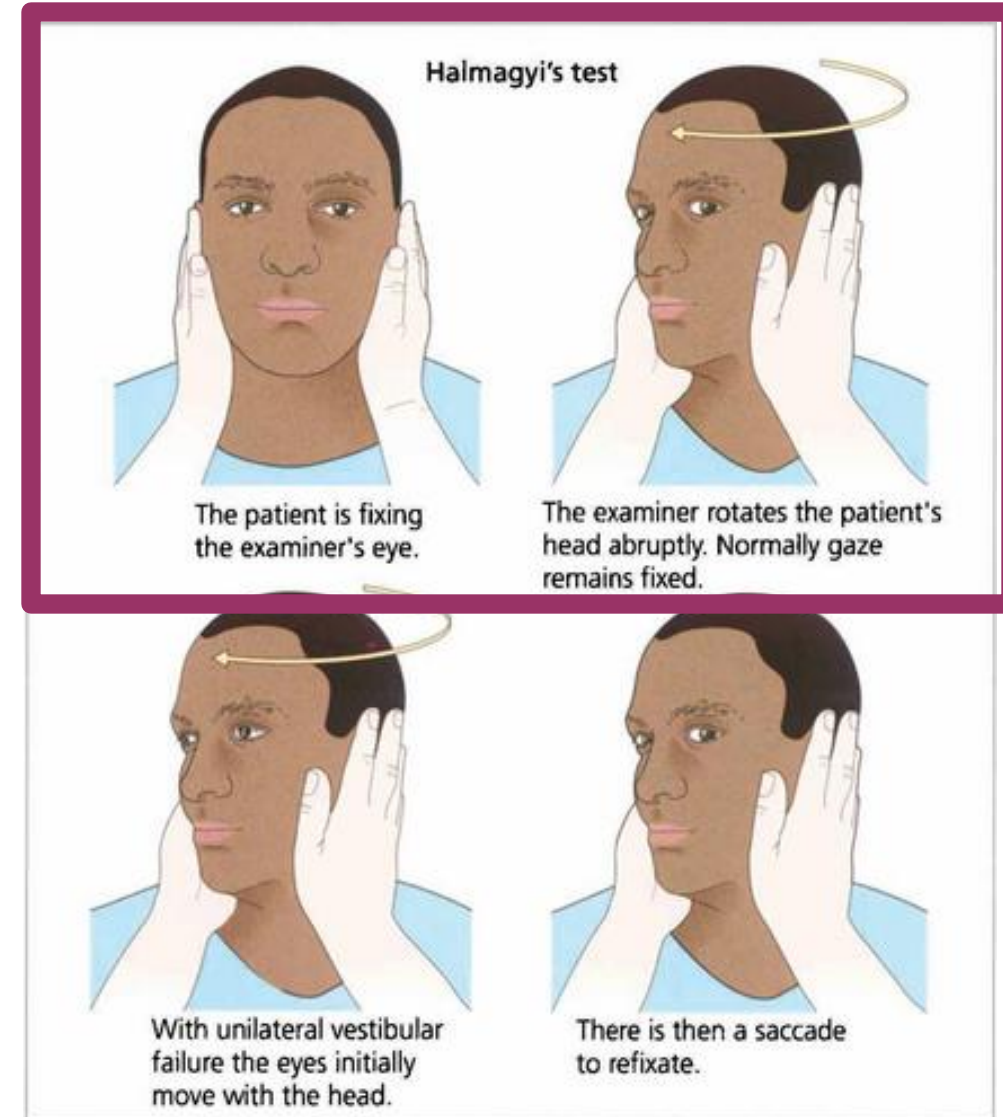
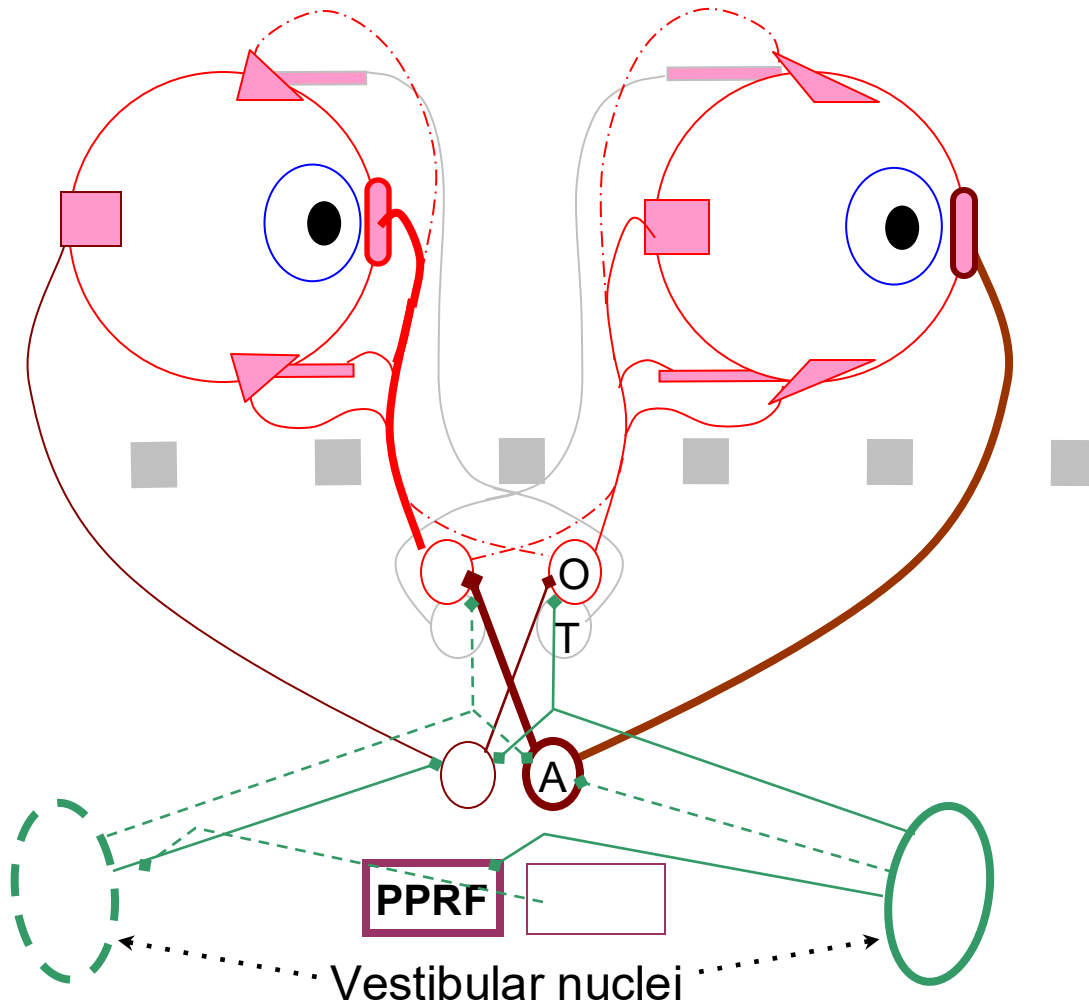
When the body is moving or when the head is moving vestibular circuits are engaged

Vestibular ipsilateral & contralateral projections to cranial nerve nuclei that control eye movement as well as projections to the PPRF



VOR- Vestibulo Occular Reflex

Maintaining eye fixation on a visual target even when head moves. Eyes move opposite head.



- Clinical Examples of Vestibular System Insults
 - Questions can be asked about these clinical presentations. If you understand the neuroanatomy and keep the pathways in mind as you are answering the questions, you will increase your chances of getting the questions right

Benign paroxysmal positional vertigo - BPPV

benign = does not worsen over time or reflect additional pathology

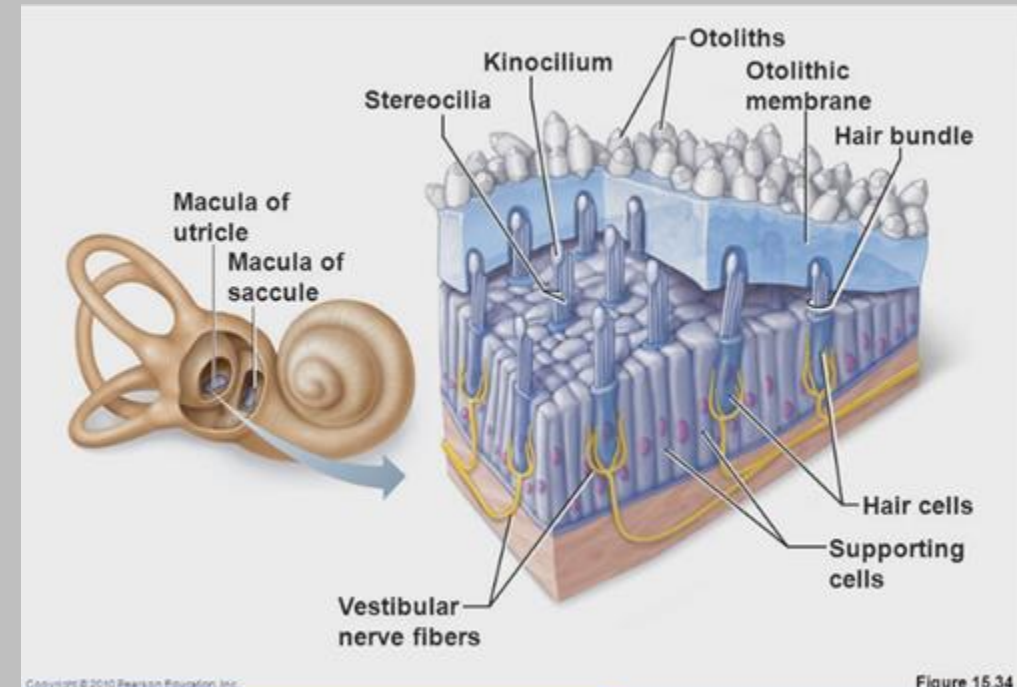
paroxysmal = abates relatively quickly (< 1 min)

positional = is elicited with particular head movements

vertigo = rotary vertigo (compelling spinning sensation) includes nystagmus and nausea

Most cases are idiopathic can be linked to head trauma or viral infection. Also, more common in older individuals and certain occupations (e.g. automechanics and plumbers).

Associated with loose otoliths that get stuck in semi-circular canals and disturb normal movement of endolymph and cristae.



BPPV testing in patients that present with vertigo: Dix-Hallpike maneuver

*Nystagmus seen during the test
= positive sign & indicative of
peripheral vestibular dysfunction
such as loose otoliths*

If test is *negative*, vertigo is likely
related to dysfunction of
vestibular circuits in central
nervous system



Video-oculography to
record eye movements

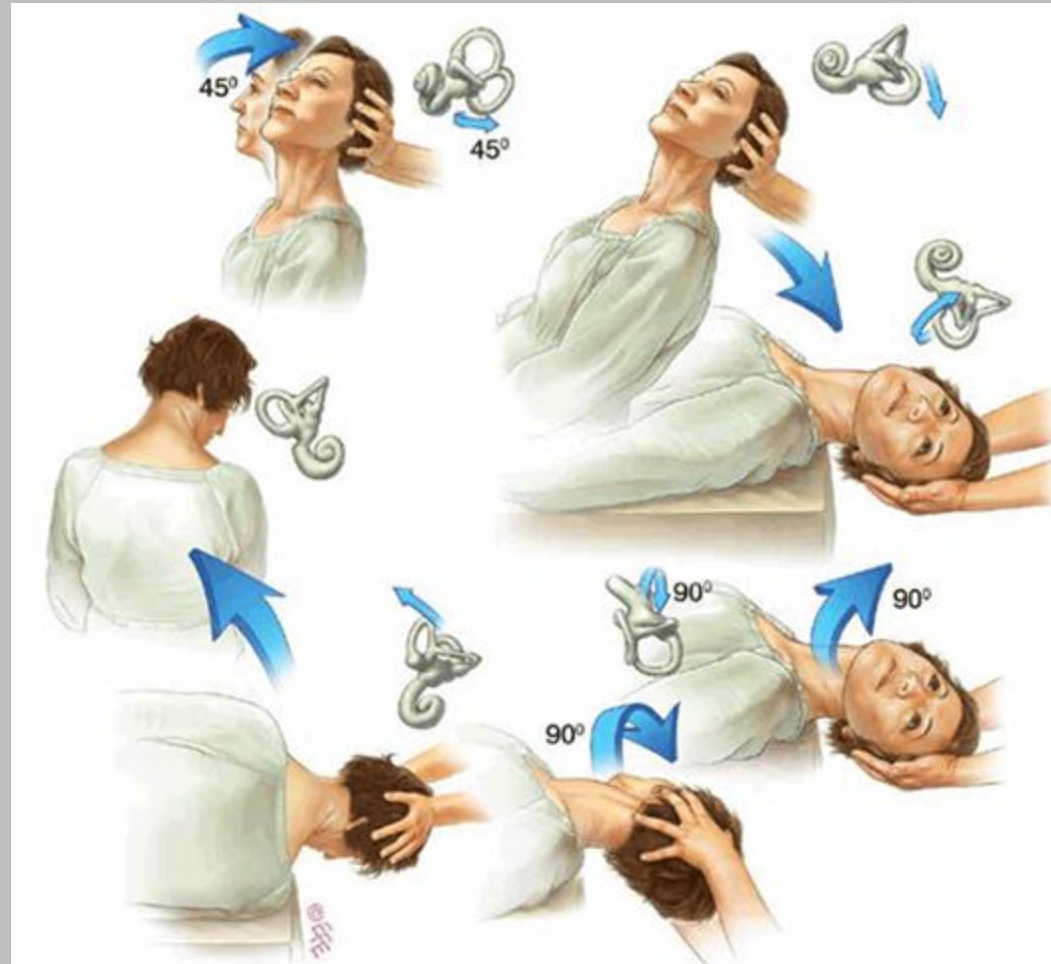


doi/full/10.1056/NEJMicm0907386

Benign paroxysmal positional vertigo treatment: Epley maneuver

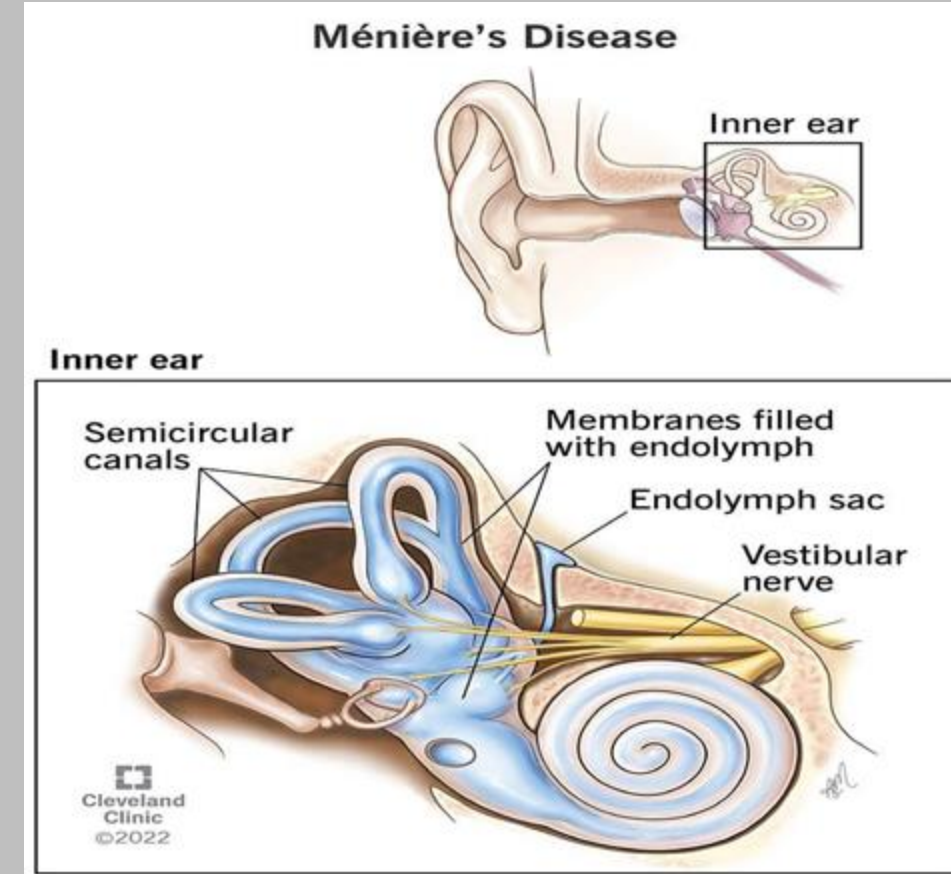
The Epley maneuver attempts to return the otoconia to the utricle.

80-90% initial success rate, 50% recurrence



Meniere Disease (PV)

- A cause of peripheral vertigo (presents worse than central vertigo)
- Caused by improper endolymph resorption in the vestibule
- Etiology: Idiopathic, but can be due to viral infections, autoimmunity. Also some genetic or environmental factors involved.
- Symptoms: Recurrent episodes of peripheral vertigo, fluctuating unilateral sensorineural hearing loss and unilateral tinnitus (Meniere triad).
- Treatment: antihistamines, benzodiazepines, anti-emetics.



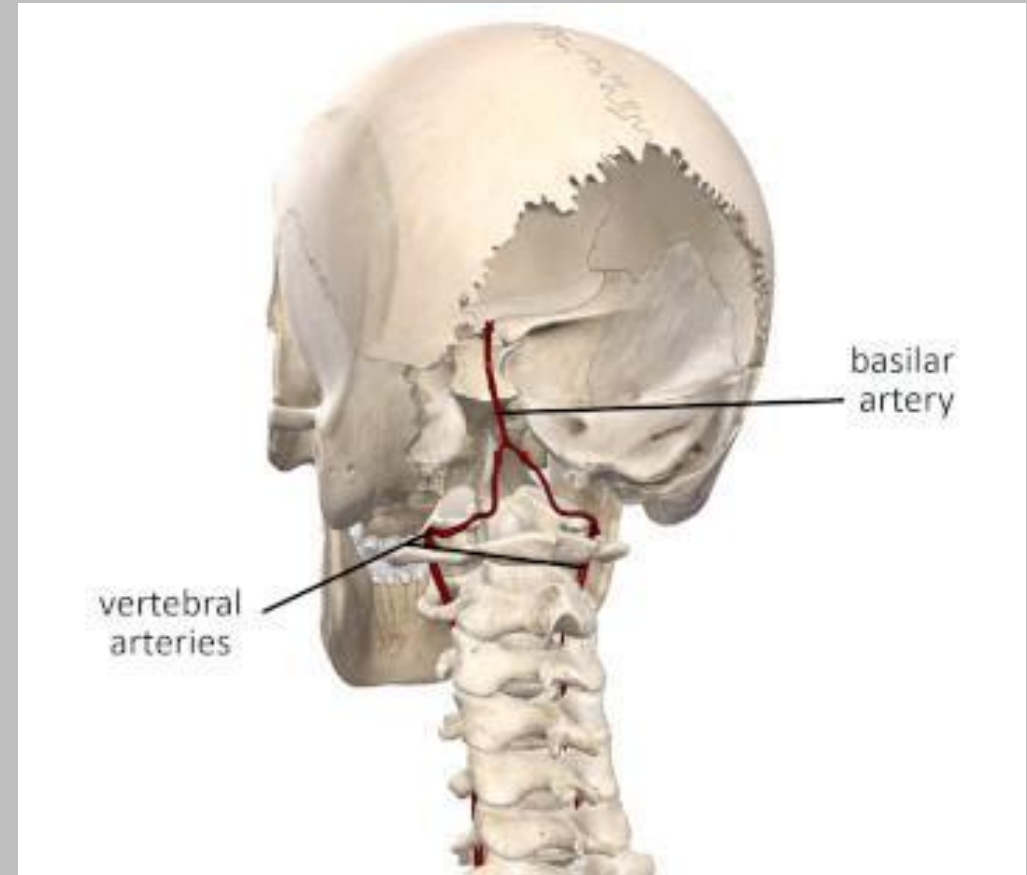
Labyrinthitis

- Caused by **Inflammation of the membranous labyrinth** of the inner ear.
 - Membranous labyrinth has endolymph which has lots of K^+ in it.
- Etiology: bacterial or viral infection, and or manifestation of systemic autoimmune disease. HIV can also cause it.
- Sx: severe vertigo, N/V, tinnitus, and/ or hearing loss.
- Usually pt makes full recovery, but some are left with residual balance or hearing issues.
- Labyrinthitis is an INFECTION of inner ear
- Meniere's is not necessarily an infection, but can be caused by infection. Meniere's is also more likely to present with multiple recurrent episodes, unlike labyrinthitis.



Vertebrobasilar Insufficiency

- Caused by inadequate blood flow through the posterior circulation. The medulla, cerebellum, pons, midbrain, thalamus and occipital cortex can all be affected in VBI. 2 vertebral arteries cross midline at C6 transverse process to become basilar artery.
- It may occur as a TIA and usually caused by atherosclerosis of vertebral arteries or basilar artery. It can also be caused by traveling embolism.
- Risk factors include: HTN, smoking, older age, male, positive family history, obesity, diabetes and hyperlipidemia
- Symptoms include: Central vertigo, syncope, nausea, visual or auditory disturbances, dysphagia and hemiparesis.



Ototoxic drugs

- Aminoglycosides
- Furosemide
- Cisplatin
- Quinine
- Aspirin/salicylates
- Erythromycin
- Vancomycin
- Opioids

- no need to memorize this yet

OTOTOXIC DRUGS

Mnemonic - "CALM EAR"

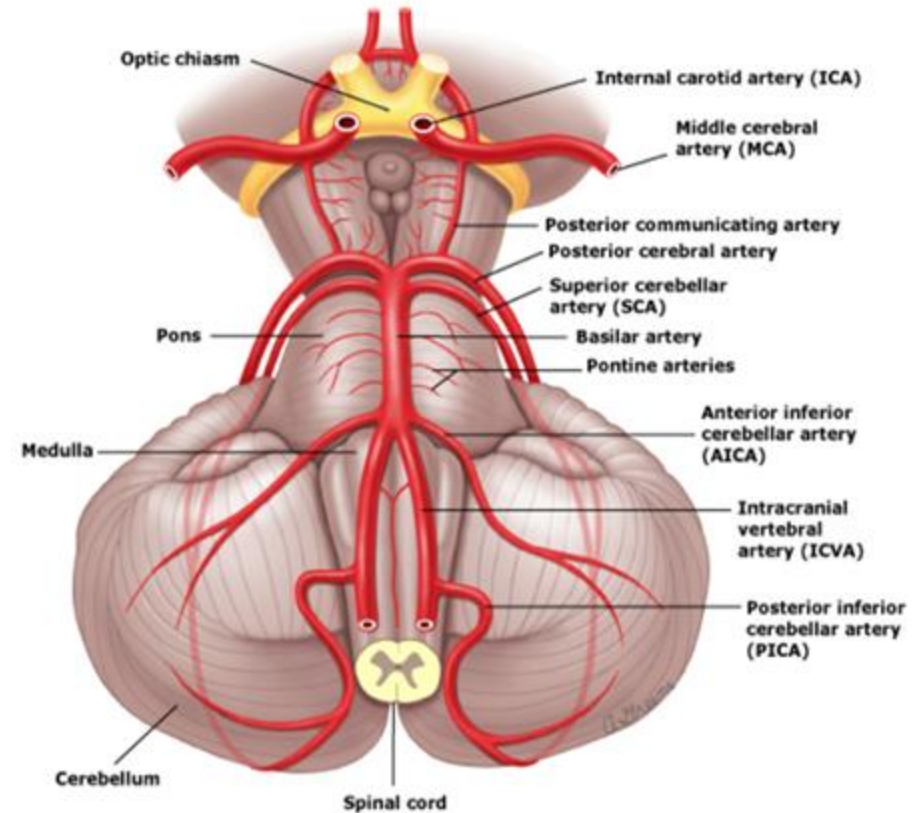
- C** - Cisplatin and Carboplatin
- A** - Aminoglycoside (Gentamycin & Tobramycin)
- L** - loop Diuretics
- M** - malaria drug-Quinine
- E** - Erythromycin
- A** - Aspirin
- R** - Redman Syndrome (vancomycin)



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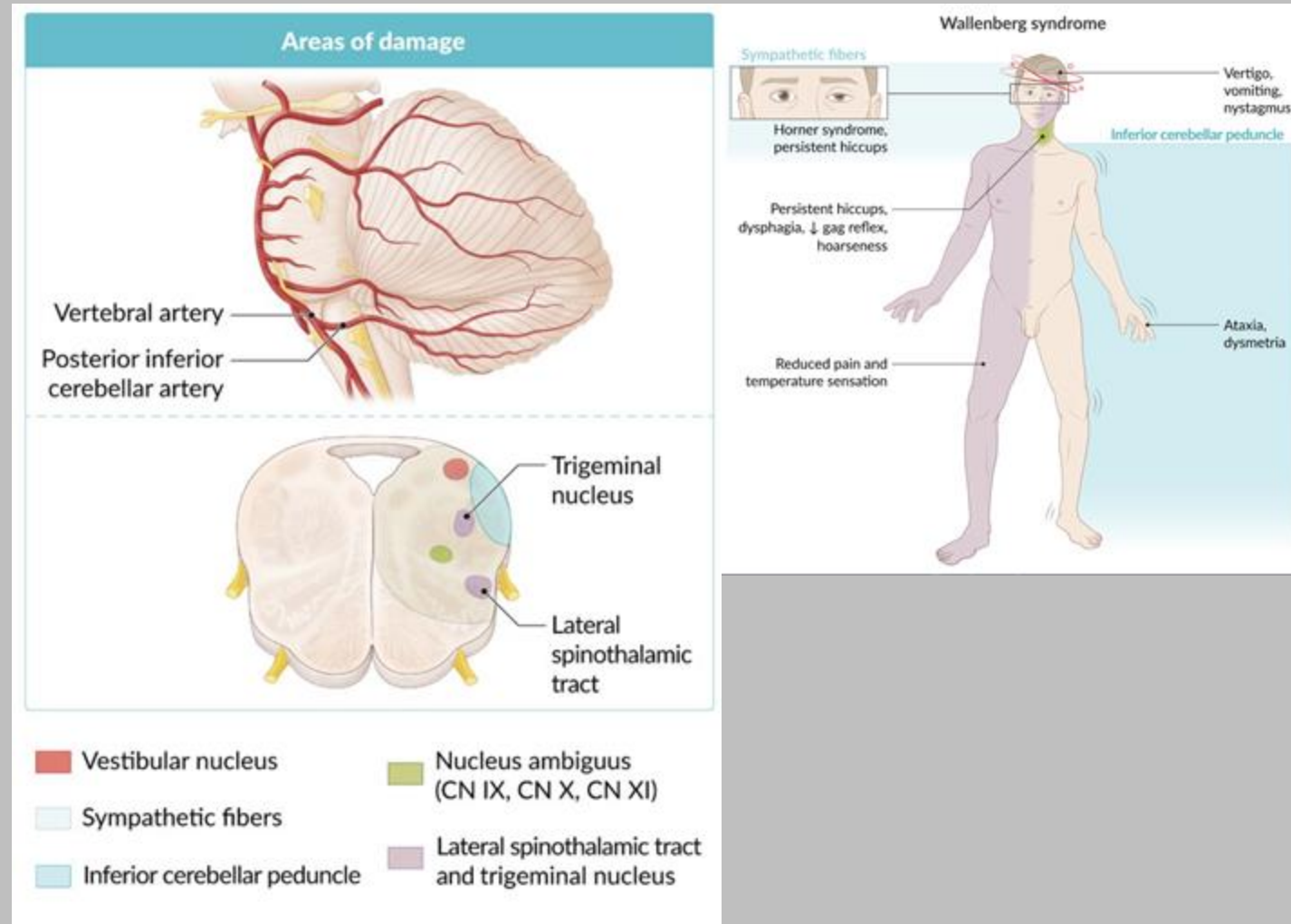
Posterior Circulation Stroke

- PCA strokes account for 20% of all strokes, most PCA strokes are ischemic→ atherosclerosis, dissection or embolism
- Most commonly occur in: Posterior cerebral artery (PCA)>posterior inferior cerebellar artery (PICA)> superior cerebellar artery (SCA)>anterior inferior cerebellar artery (AICA).
- Pt can present with central vertigo from vestibular nuclear involvement at pontomedullary junction
- Ataxia (cerebellar involvement)
- Nystagmus
- Vomiting
- Headache
- Motor weakness



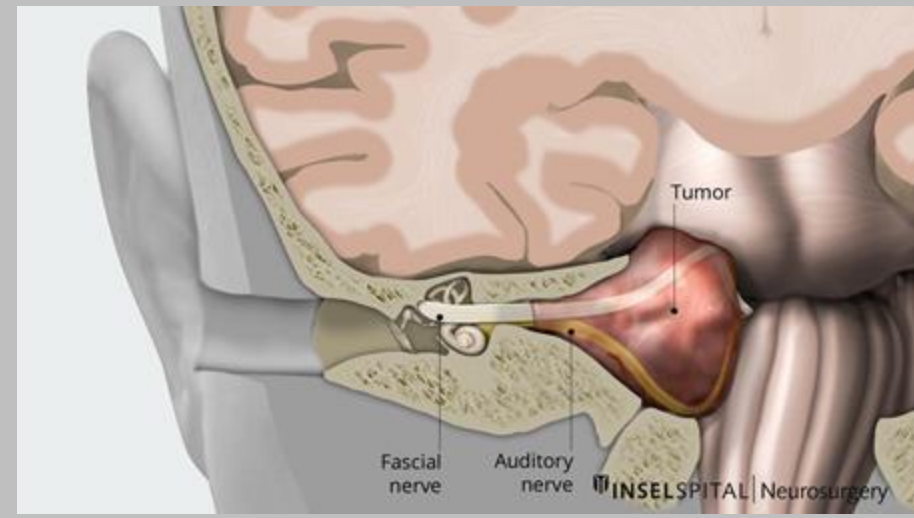
Lateral Medullary /Wallenberg Syndrome (CV)

- A cause of central vertigo.
- Caused by occlusion of PICA.
- Causes an Ipsilateral
 - vertigo and nystagmus (central, beats in direction of gaze) Vestibular nucleus damage
 - ataxia and dysmetria from inferior cerebellar peduncle insult, carry signal from bipolar cells from vestibular ganglion to cerebellum and brainstem.
 - Dysphagia, hoarseness and dysphonia from nucleus ambiguus insult
 - Horner Syndrome and hiccups from insulting sympathetic fibers in area.
- Contralateral:
 - loss of pain and temp in trunks and limb (LSTT)

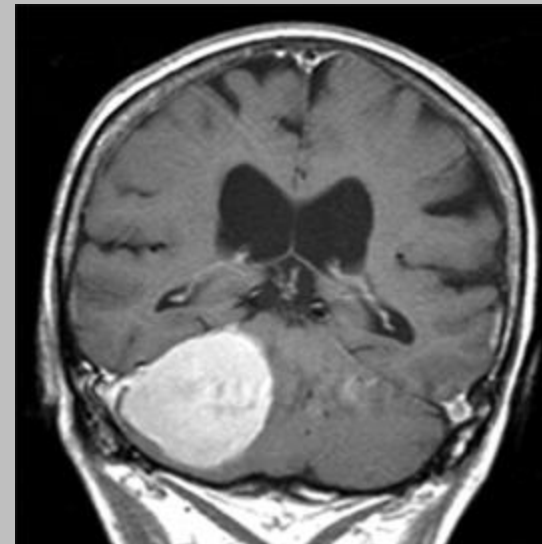


Posterior Fossa Tumors

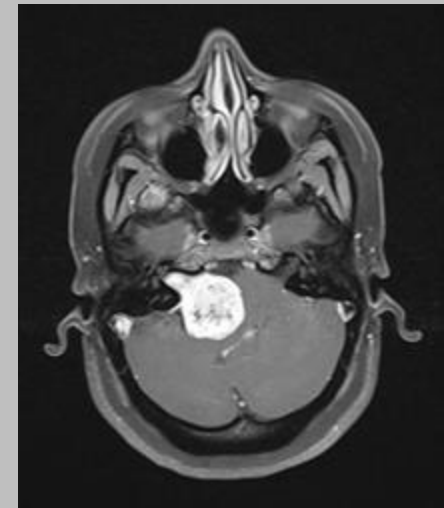
- Vestibular Schwannoma, (CV/PV)
 - Benign tumor that arises from the Schwann cells of the CN VIII.
 - Often also involves CN VII due to proximity: bells palsy. CN 7 leaves through stylomastoid foramen and CN8 and Cn7 run very close together. So a VS can also cause CNVII issues.
- Posterior fossa meningiomas
 - Brain tumors that arise from the meningeal layer.
 - Compression of vestibular nuclei, flocculonodular lobe, cerebellum and other vestibular pathway.



Right inferior
tentorial
meningioma



Vestibular
Schwannoma
w/cerebellar
compression



Multiple Sclerosis

- Chronic demyelination of CNS caused by an autoimmunity reaction towards oligodendrocytes of the CNS.
- Can lead to demyelination of cerebellar and vestibular neurons and their axons.

Central vs Peripheral Vertigo

Central Vertigo	Peripheral Vertigo
Gradual onset	Sudden onset
Mild intensity	Severe intensity
Mild nausea/vomiting	Intense nausea/vomiting
Associated neurologic findings typically present	No associated neurologic findings
Mild nystagmus	Relatively intense nystagmus
Position change has mild effect	Position change has intense effect
No refractoriness—can repeat the “tilt” test and patient responds every time	Rapidly refractory—cannot repeat the “tilt” test; patient will not respond again
Patient falls to same side as lesion	Patient falls to same side as lesion

Points of emphasis

- Vestibular end organ anatomy & physiology (hair cells) underlies sensory transduction and perception of our bodies moving in the environment
- Pathways from vestibular nuclei underlie perceptual and behavioral reflex output
- Pathologies of vestibular system can impair orientation (vertigo), balance, and eye movements.

Questions

A 45 year old man presents with “dizziness” for the past month. The pt also feels he has a loss of coordination of limb movements on the right side of his body and feels like he wants to fall to his right side. The pt also finished a course of aminoglycosides to treat a *Pseudomonas aeruginosa* infection. The Pt also complains of loss of pain and temperature on the right side of his face and well as the left side of his body. On exam the pt has a horizontal nystagmus that beats when the patient looks to the left and the right. What is the most likely diagnosis?

- A) BPPV
- B) MCA stroke
- C) Lateral Medullary Syndrome (Wallenberg Syndrome)
- D) Drug induced ototoxicity
- E) Meniere's disease
- F) Labyrinthitis

Answer

Lateral Medullary Syndrome (Wallenberg Syndrome)

Feedback is welcomed

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

